

CET104 – UFCD5106

Final Project Computer Networks – Implementation of NETCORP Corporate Network



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Introduction

This project implements a secure and redundant enterprise network for the NetCorp enterprise on Cisco Packet Tracer.

The network connects a Data Center to three branches, ensuring internal communication, Internet access and centralized service management. Topology applies routing, switching, and security concepts learned in the course.

The solution combines VLANs, HSRP, multiple routing protocols (OSPF, EIGRP, RIP) with redistribution, NAT, ACLs, and AAA/802.1X authentication.

The entire configuration was validated through connectivity and failover testing.

Objectives

- Implement secure switching with VLANs, VTP, EtherChannel and Layer 2 security (Port Security, DHCP Snooping).
- Ensure redundancy with HSRP in the Data Center and dynamic routing (OSPF, EIGRP, RIP) with redistribution.
- Ensure security with ACLs, NAT, AAA/802.1X authentication, and SSH.
- Provide core services (DHCP, DNS, VPN) via SRV1 server.
- Validate and document the implementation with extensive testing.

Statement / Exercises

The NetCorp company has expanded its operations and now has a Data Center and three interconnected branches.

Network technicians must implement the communication and security infrastructure that supports internal services, Internet access, and management policies.

The topology includes:

- Two routers in the data center (R1 and R2) – Responsible for redundancy and route redistribution;
- Three branches (A, B and C) with their own routers (R3, R4 and R5) and access switches;
- Central Service (SRV1) with DHCP, DNS, TACACS+ and Syslog;
- Internet network (ISP) for external connection.

Rede	Endereço	Máscara	Função
Backbone R1–R2	10.0.0.0/30	/30	HSRP
Filial A	10.0.1.0/24	/24	OSPF
Filial B	10.0.2.0/24	/24	EIGRP
Filial C	10.0.3.0/24	/24	RIP
VLAN 10	192.168.10.0/26	/26	Administração
VLAN 20	192.168.20.0/26	/26	Vendas
VLAN 30	192.168.30.0/26	/26	Financeiro
VLAN 99	192.168.99.0/30	/30	Gestão
Rede ISP	200.10.10.0/30	/30	Internet

Setup steps:

- 1) VLANs, VTP and Etherchannel
 - Create VLANs 10 (Admin), 20 (Sales), 30 (Financial), and 99 (Management).
 - Configure SW1 as VTP Server and the remaining switches as VTP Clients.
 - Create EtherChannel (LACP) between SW1 and SW2.
- 2) Redundancy with HSRP
 - Implement HSRP between routers R1 and R2.
 - R1 → priority 110 (active).
 - R2 → priority 100 (standby).
 - Test failover by shutting down the primary router.
- 3) Dynamic Routing and Redistribution
 - R3 (Branch A) → OSPF Area 0.
 - R4 (Branch B) → EIGRP AS 100.
 - R5 (Branch C) → RIP version 2.
 - R1 or R2 must redistribute the routes between OSPF, EIGRP, and RIP.

4) NAT (PAT) on all routers

- Implement NAT overload (PAT) on all routers (R1 to R5).
- LAN interfaces should be inside, and interfaces to the ISP outside.
- Confirm internet access (ping to the ISP's loopback or IP interface).

5) ACLs – Access Control Lists

- Server Access (R1) – Only VLAN 10 can access the TACACS+ server and DNS.
- Inter-VLAN Restriction – Prevent direct communication between VLAN 20 and VLAN 30.
- Management (SSH) – Only VLAN 10 can access the equipment via SSH.

6) Layer 2 Security

- Port-Security – Apply Port-Security to all PC access ports, it should allow only one mac-address, it should be learned dynamically and if any violation occurs, it should immediately disconnect the port.
- DHCP Snooping – Enable DHCP Snooping, mark ports as trusted and untrusted.
- Dynamic ARP Inspection – Enable DAI using the DHCP Snooping database.

7) AAA, IEEE 802.1X and SSH authentication

- Configure centralized authentication via TACACS+/RADIUS on the SRV1 server.
- Enable the port-based network access control protocol by configuring an AAA server to decide whether a device can access the network. (Users of your choice, password cinel).
- Enable SSH on all equipment and disable telnet.

8) VPN – Virtual Private Network

- One of the machines on a LAN must access another network via a remote access VPN.

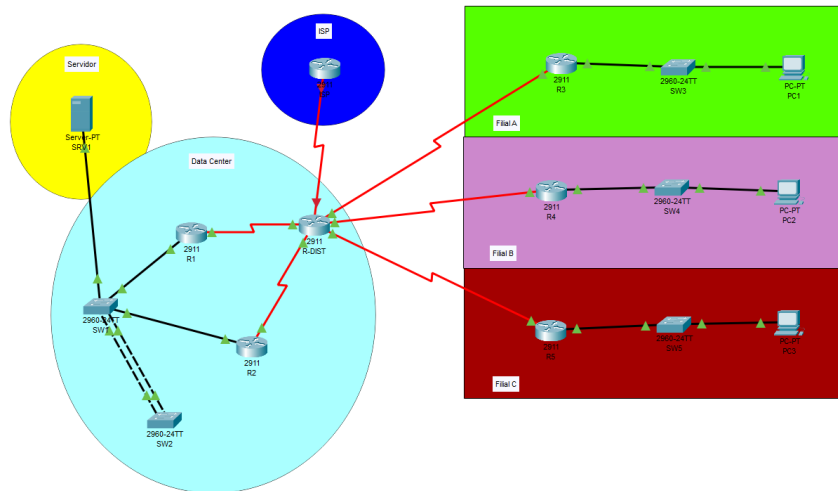
9) Testing and Validation

- Test ping between branch offices.
- Check route redistribution with the show ip route command.
- Validate NAT with the show ip nat translations command.
- Confirm ACLs and port security (show access-lists, show port-security).
- Test AAA and SSH.

To save the settings I perform the "wr" command periodically.

Topology

Since this work offers more freedom when creating the topology, I decided to go with this star format.

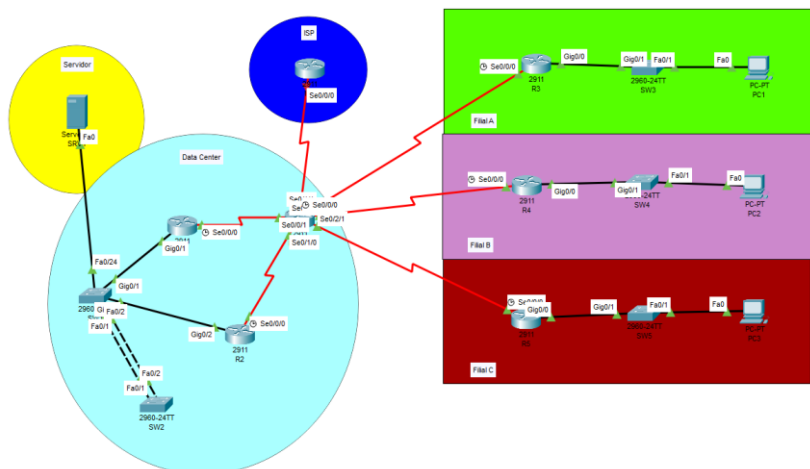


I decided to implement a central router "R-DIST" to centralize redistribution and simplify route management. In each exercise I will add the necessary settings for this router to work properly.

Key points of the topology:

- R-DIST is the center hub (star)
- All routers (R1, R2, R3, R4, R5) connect to R-DIST
- Data Center has R1 and R2 both connected to SW1
- SW2 only works for EtherChannel with SW1 (does not connect to routers)
- Each branch has its router connected directly to R-DIST

The topology will have these connections:

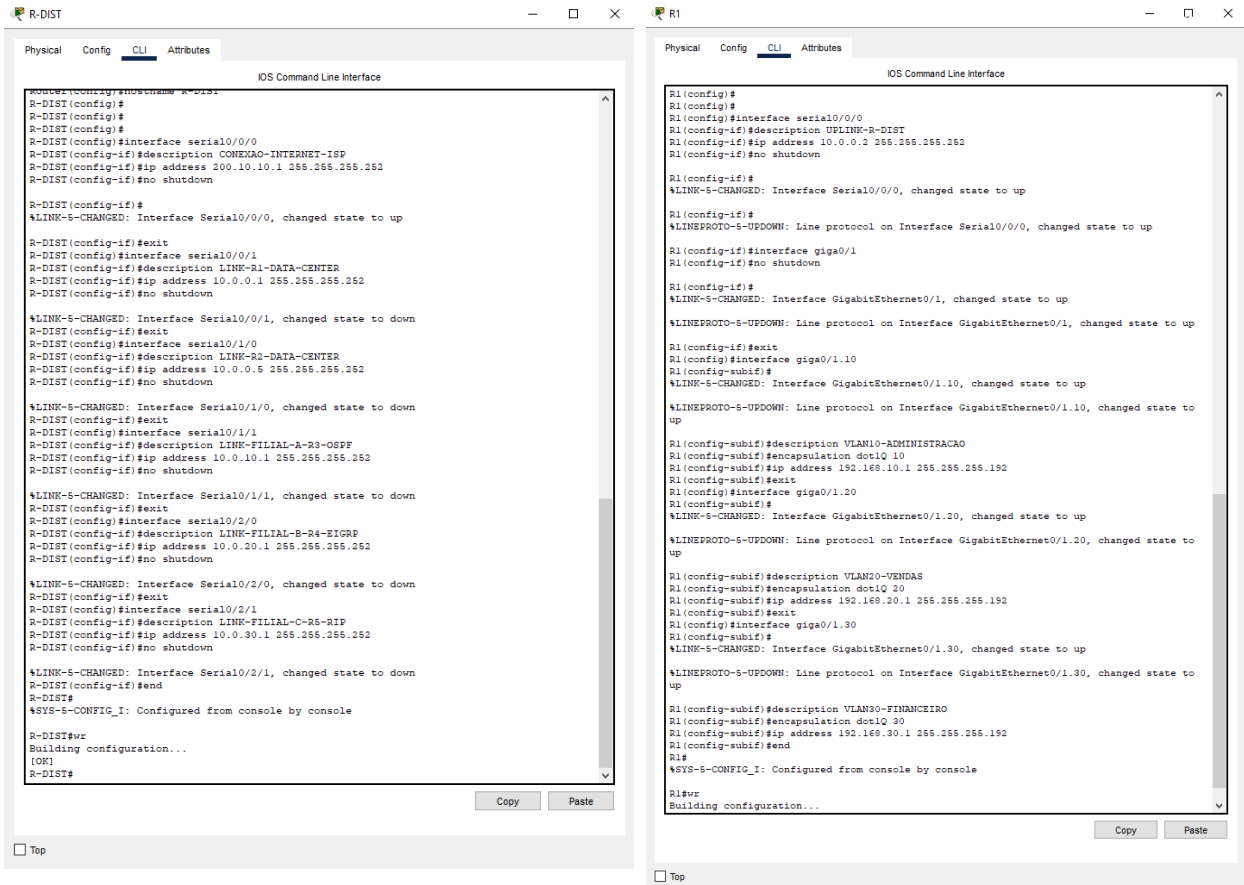


Additionally I will edit the hostname of each device to be the same as the names in the topology.

IP Addressing

I will now assign IPs according to the table present in the statement.

R-DIST: R1:



```
Router(config)#hostname R-DIST
R-DIST(config)#
R-DIST(config)#
R-DIST(config)#interface serial0/0/0
R-DIST(config-if)#description CONEXAO-INTERNET-ISP
R-DIST(config-if)#ip address 200.10.10.1 255.255.255.252
R-DIST(config-if)#no shutdown

R-DIST(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R-DIST(config-if)#exit
R-DIST(config)#interface serial0/0/1
R-DIST(config-if)#description LINK-R2-DATA-CENTER
R-DIST(config-if)#ip address 10.0.0.1 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/1/0
R-DIST(config-if)#description LINK-R2-DATA-CENTER
R-DIST(config-if)#ip address 10.0.0.5 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/1/1
R-DIST(config-if)#description LINK-FILIAL-A-R3-OSPF
R-DIST(config-if)#ip address 10.0.10.1 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/2/0
R-DIST(config-if)#description LINK-FILIAL-B-R4-EIGRP
R-DIST(config-if)#ip address 10.0.20.1 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/2/1
R-DIST(config-if)#description LINK-FILIAL-C-R5-RIP
R-DIST(config-if)#ip address 10.0.30.1 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/1, changed state to down
R-DIST(config-if)#end
R-DIST#
%SYS-5-CONFIG_I: Configured from console by console

R-DIST#wr
Building configuration...
[OK]
R-DIST#
```

```
R1(config)#
R1(config)#interface serial0/0/0
R1(config-if)#description UPLINK-R-DIST
R1(config-if)#ip address 10.0.0.2 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R1(config-if)#interface giga0/1
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#interface giga0/1.10
R1(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.10, changed state to up

R1(config-subif)#description VLAN10-ADMINISTRACAO
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.192
R1(config-subif)#exit
R1(config)#interface giga0/1.20
R1(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.20, changed state to up

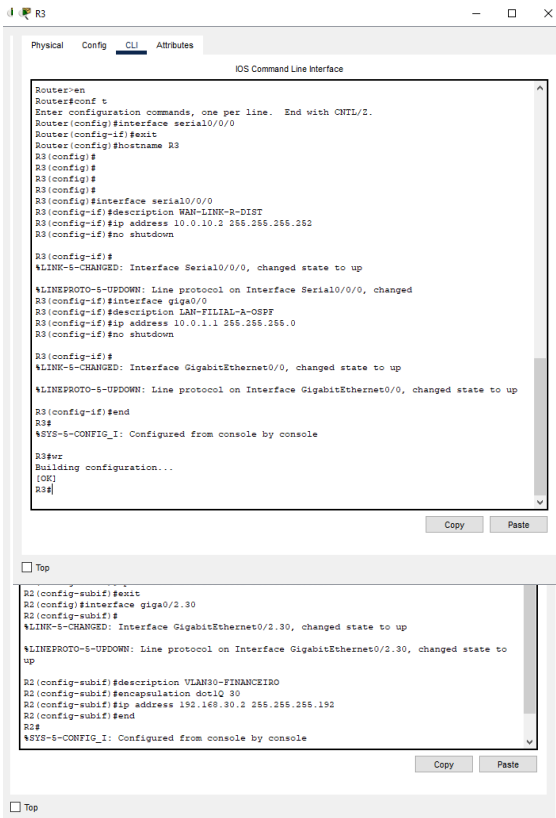
R1(config-subif)#description VLAN20-VENDAS
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.192
R1(config-subif)#exit
R1(config)#interface giga0/1.30
R1(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to up

R1(config-subif)#description VLAN30-FINANCEIRO
R1(config-subif)#encapsulation dot1Q 30
R1(config-subif)#ip address 192.168.30.1 255.255.255.192
R1(config-subif)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#wr
Building configuration...
```

R-DIST: Configuration of IP addresses on WAN interfaces (ISP, R1, R2) and links to branches R3, R4 and R5.

R2: R3:



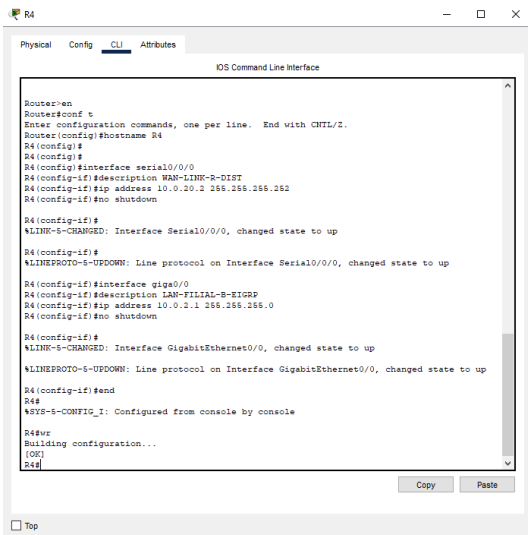
```
Router>en
Router>conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial0/0/0
Router(config-if)#exit
Router(config)#hostname R3
R3(config)#
R3(config)#
R3(config)#
R3(config)#interface serial0/0/0
R3(config-if)#description WAN-LINK-R-DIST
R3(config-if)#ip address 10.0.10.2 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
R3(config-if)#interface giga0/0
R3(config-if)#description LAN-FILIAL-A-OSPF
R3(config-if)#ip address 10.0.1.1 255.255.255.0
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R3(config-if)#end
R3#
$SYS-5-CONFIG_I: Configured from console by console

R3#vr
Building configuration...
[OK]
R3#
```

R4:R5:



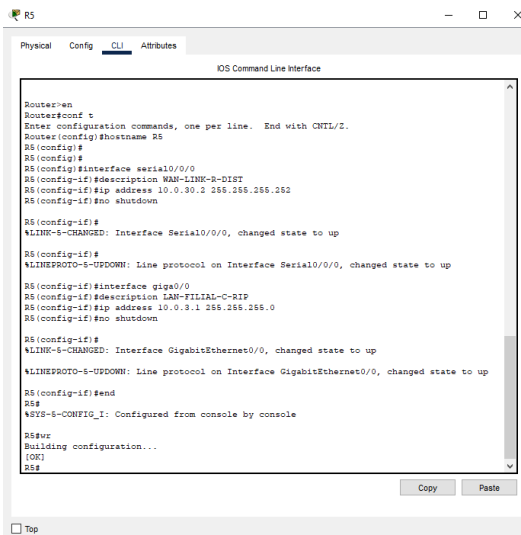
```
Router>en
Router>conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#
R4(config)#
R4(config)#interface serial0/0/0
R4(config-if)#description WAN-LINK-R-DIST
R4(config-if)#ip address 10.0.20.2 255.255.255.252
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
R4(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R4(config-if)#interface giga0/0
R4(config-if)#description LAN-FILIAL-B-RIP
R4(config-if)#ip address 10.0.2.1 255.255.255.0
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R4(config-if)#end
R4#
$SYS-5-CONFIG_I: Configured from console by console

R4#vr
Building configuration...
[OK]
R4#
```



```
Router>en
Router>conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R5
R5(config)#
R5(config)#
R5(config)#interface serial0/0/0
R5(config-if)#description WAN-LINK-R-DIST
R5(config-if)#ip address 10.0.30.2 255.255.255.252
R5(config-if)#no shutdown

R5(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
R5(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R5(config-if)#interface giga0/0
R5(config-if)#description LAN-FILIAL-C-RIP
R5(config-if)#ip address 10.0.3.1 255.255.255.0
R5(config-if)#no shutdown

R5(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R5(config-if)#end
R5#
$SYS-5-CONFIG_I: Configured from console by console

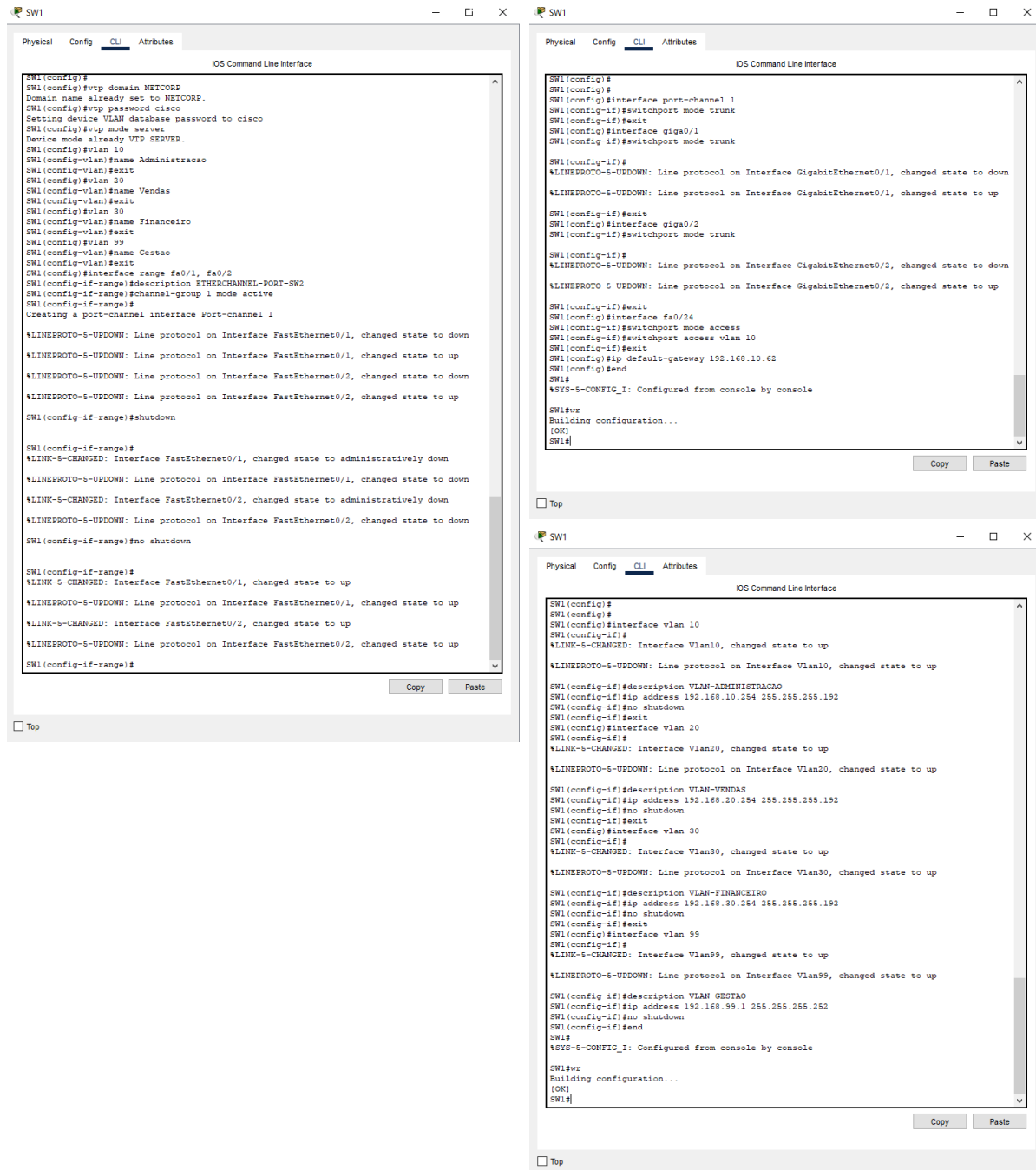
R5#vr
Building configuration...
[OK]
R5#
```

R1 and R2: Assignment of IPs on the interfaces connecting to R-DIST and on the subinterfaces of VLANs 10, 20 and 30.

R3, R4, R5: Configuration of IP addresses on WAN (connection to R-DIST) and LAN (internal branch networks) interfaces.

1) VLANs, VTP and Etherchannel

SW1:



```
SW1 (config)#
SW1(config)#vtp domain NETCORP
Domain name already set to NETCORP.
SW1(config)#vtp password cisco
Setting device VLAN database password to cisco
SW1(config)#vtp mode server
Device mode already VTP SERVER.
SW1(config)#vlan 10
SW1(config-vlan)#name Administracao
SW1(config-vlan)#exit
SW1(config)#vlan 20
SW1(config-vlan)#name Vendas
SW1(config-vlan)#exit
SW1(config)#vlan 30
SW1(config-vlan)#name Financeiro
SW1(config-vlan)#exit
SW1(config)#vlan 99
SW1(config-vlan)#name Gestao
SW1(config-vlan)#exit
SW1(config)#interface range fa0/1, fa0/2
SW1(config-if-range)#description ETHERCHANNEL-PORT-SW2
SW1(config-if-range)#channel-group 1 mode active
SW1(config-if-range)#
Creating a port-channel interface Port-channel 1
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
SW1(config-if-range)#shutdown
SW1(config-if-range)#
%LINK-6-CHANGED: Interface FastEthernet0/1, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-6-CHANGED: Interface FastEthernet0/2, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
SW1(config-if-range)#no shutdown
SW1(config-if-range)#
%LINK-6-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-6-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
SW1(config-if-range)#

SW1 (config)#
SW1(config)#interface port-channel 1
SW1(config-if)#switchport mode trunk
SW1(config-if)#exit
SW1(config)#interface giga0/1
SW1(config-if)#switchport mode trunk

SW1 (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
SW1 (config-if)#exit
SW1 (config)#interface giga0/2
SW1 (config-if)#switchport mode trunk

SW1 (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
SW1 (config-if)#exit
SW1 (config)#interface fa0/24
SW1 (config-if)#switchport mode access
SW1 (config-if)#switchport access vlan 10
SW1 (config-if)#exit
SW1 (config)#ip default-gateway 192.168.10.62
SW1 (config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#vr
Building configuration...
[OK]
SW1#
```

```
SW1 (config)#
SW1(config)#interface vlan 10
SW1(config-if)#
%LINK-6-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

SW1 (config-if)#description VLAN-ADMINISTRACAO
SW1 (config-if)#ip address 192.168.10.254 255.255.255.192
SW1 (config-if)#no shutdown
SW1 (config-if)#exit
SW1 (config)#interface vlan 20
SW1 (config-if)#
%LINK-6-CHANGED: Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

SW1 (config-if)#description VLAN-VENDAS
SW1 (config-if)#ip address 192.168.20.254 255.255.255.192
SW1 (config-if)#no shutdown
SW1 (config-if)#exit
SW1 (config)#interface vlan 30
SW1 (config-if)#
%LINK-6-CHANGED: Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up

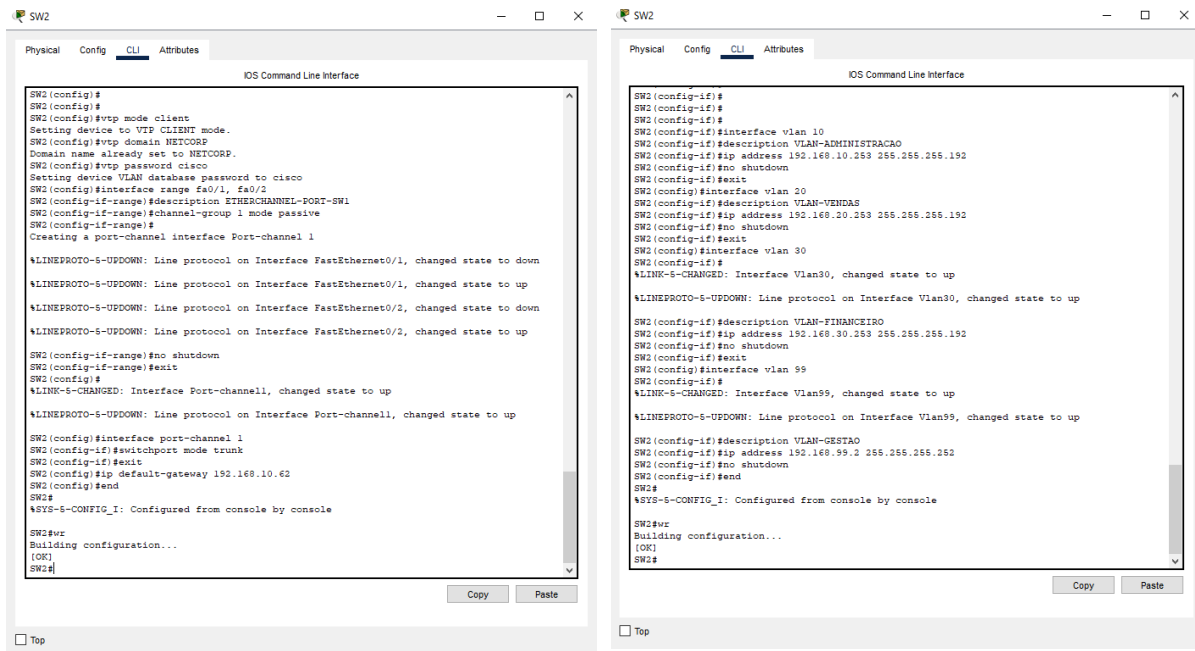
SW1 (config-if)#description VLAN-FINANCEIRO
SW1 (config-if)#ip address 192.168.30.254 255.255.255.192
SW1 (config-if)#no shutdown
SW1 (config-if)#exit
SW1 (config)#interface vlan 99
SW1 (config-if)#
%LINK-6-CHANGED: Interface Vlan99, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up

SW1 (config-if)#description VLAN-GESTAO
SW1 (config-if)#ip address 192.169.99.1 255.255.255.252
SW1 (config-if)#no shutdown
SW1 (config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#vr
Building configuration...
[OK]
SW1#
```

SW1 (VTP Server): Creation of VLANs 10, 20, 30 and 99. VTP Server activation and configuration of EtherChannel with LACP for SW2.

SW2:



```
SW2 (config)#
SW2 (config)#vtp mode client
Setting device to VTP CLIENT mode.
SW2 (config)#vtp domain NETCORP
Domain name already set to NETCORP.
SW2 (config)#vtp password cisco
Setting device VLAN database password to cisco
SW2 (config)#interface range fa0/1, fa0/2
SW2 (config-if-range)#description ETHERCHANNEL-PORT-SW1
SW2 (config-if-range)#channel-group 1 mode passive
SW2 (config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
SW2 (config-if-range)#no shutdown
SW2 (config-if-range)#exit
SW2 (config)#
%LINK-6-CHANGED: Interface Port-channel1, changed state to up

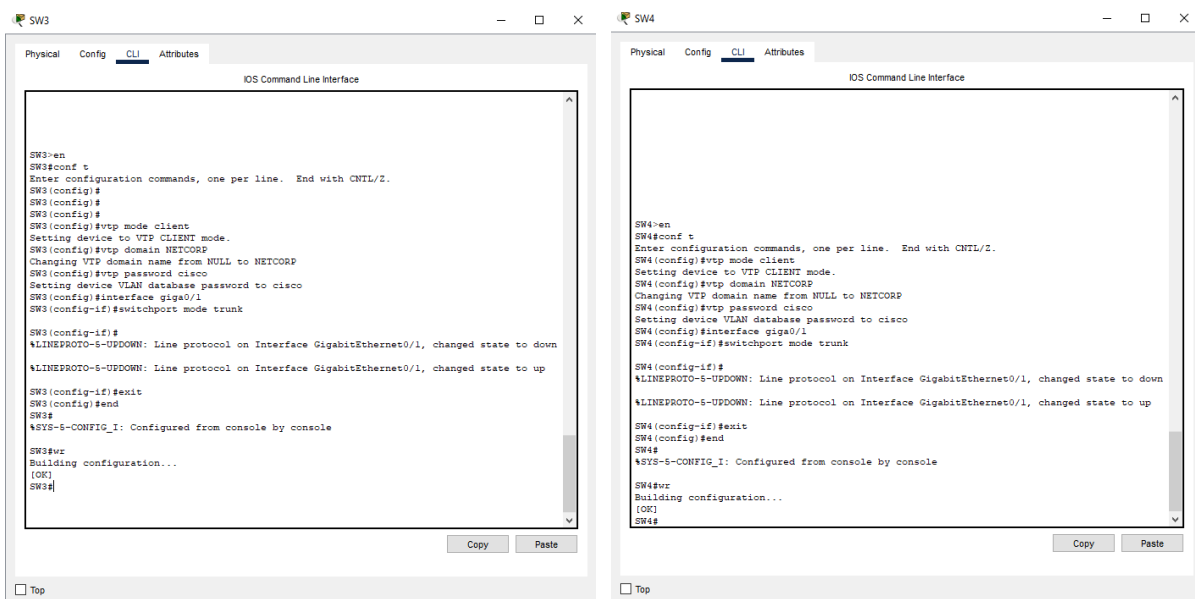
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

SW2 (config)#interface port-channel 1
SW2 (config-if)#switchport mode trunk
SW2 (config-if)#exit
SW2 (config)#ip default-gateway 192.168.10.62
SW2 (config)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console

SW2#vr
Building configuration...
[OK]
SW2#
```

SW2 (VTP Client): Configuration as VTP Client and participation in EtherChannel with SW1.

SW3:SW4:



```
SW3#en
SW3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW3 (config)#
SW3 (config)#
SW3 (config)#vtp mode client
Setting device to VTP CLIENT mode.
SW3 (config)#vtp domain NETCORP
Changing VTP domain name from NULL to NETCORP
SW3 (config)#vtp password cisco
Setting device VLAN database password to cisco
SW3 (config)#interface giga0/1
SW3 (config-if)#switchport mode trunk

SW3 (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
SW3 (config-if)#exit
SW3 (config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

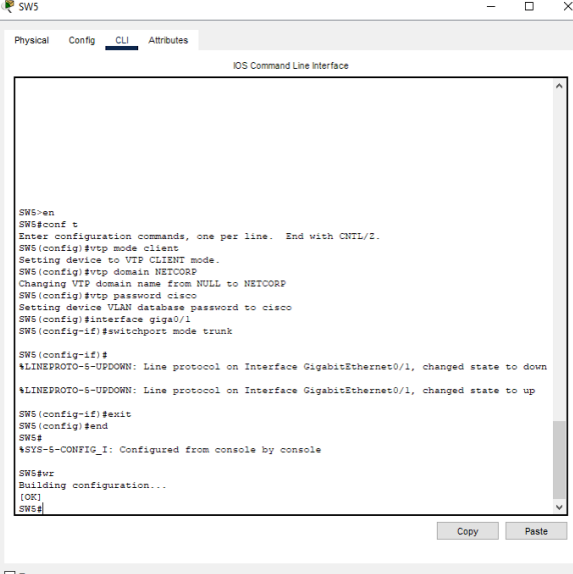
SW3#vr
Building configuration...
[OK]
SW3#

SW4#en
SW4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW4 (config)#vtp mode client
Setting device to VTP CLIENT mode.
SW4 (config)#vtp domain NETCORP
Changing VTP domain name from NULL to NETCORP
SW4 (config)#vtp password cisco
Setting device VLAN database password to cisco
SW4 (config)#interface giga0/1
SW4 (config-if)#switchport mode trunk

SW4 (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
SW4 (config-if)#exit
SW4 (config)#end
SW4#
%SYS-5-CONFIG_I: Configured from console by console

SW4#vr
Building configuration...
[OK]
SW4#
```

SW5:



```
SW5>en
SW5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW5(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW5(config)#vtp domain NETCORP
Changing VTP domain name from NULL to NETCORP
SW5(config)#vtp password cisco
Setting device VLAN database password to cisco
SW5(config)#interface giga0/1
SW5(config-if)#switchport mode trunk

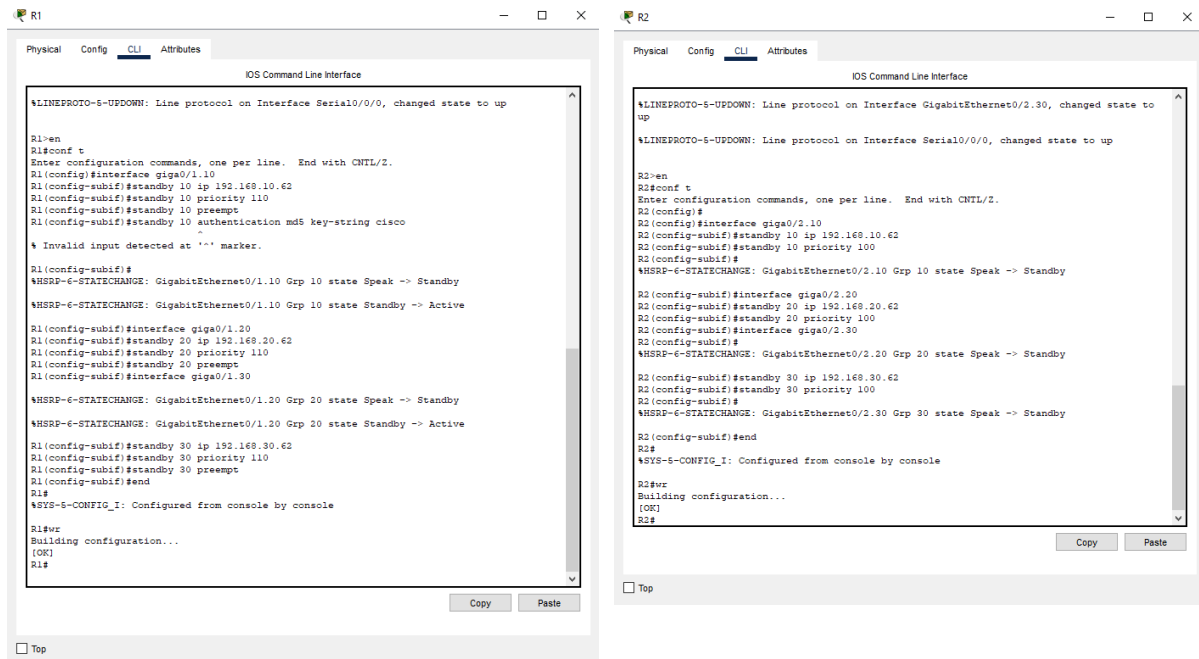
SW5(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
SW5(config-if)#exit
SW5(config)#end
SW5#
%SYS-5-CONFIG_I: Configured from console by console

SW5#sh
Building configuration...
(OX)
SW5#
```

SW3, SW4, SW5: Configuration as VTP Clients and activation of trunk ports for branch routers.

2) Redundancy with HSRP

R1:R2:



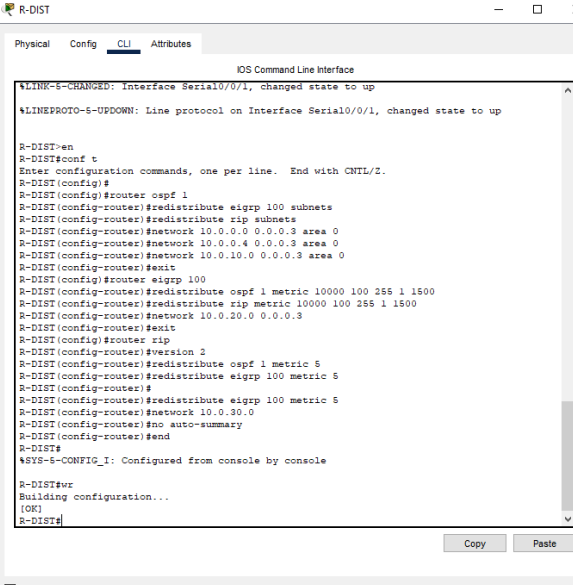
The image shows two side-by-side screenshots of the Cisco IOS Command Line Interface (CLI) for routers R1 and R2. Both windows have tabs for Physical, Config, CLI, and Attributes, with the CLI tab selected. The R1 window shows the configuration for R1, where HSRP is configured on interfaces GigabitEthernet0/1.10, GigabitEthernet0/1.20, and GigabitEthernet0/30. R1 is configured as the active gateway with a priority of 110 and the preempt command. The R2 window shows the configuration for R2, where HSRP is configured on interfaces GigabitEthernet0/2.10, GigabitEthernet0/2.20, and GigabitEthernet0/2.30. R2 is configured as the standby gateway with a priority of 100. Both routers show state change messages indicating that R1 is in the 'Speak' state and R2 is in the 'Standby' state for all three groups. The configuration commands for R1 include: `R1>en`, `R1#conf t`, `R1(config)#interface giga0/1.10`, `R1(config-subif)#standby 10 ip 192.168.10.62`, `R1(config-subif)#standby 10 priority 110`, `R1(config-subif)#standby 10 preempt`, `R1(config-subif)#standby 10 authentication md5 key-string cisco`, `R1(config-subif)#`, `R1(config-subif)#interface giga0/1.20`, `R1(config-subif)#standby 20 ip 192.168.20.62`, `R1(config-subif)#standby 20 priority 110`, `R1(config-subif)#standby 20 preempt`, `R1(config-subif)#interface giga0/1.30`, `R1(config-subif)#standby 30 ip 192.168.30.62`, `R1(config-subif)#standby 30 priority 110`, `R1(config-subif)#standby 30 preempt`, `R1(config-subif)#end`, `R1#`, `%SYS-5-CONFIG_I: Configured from console by console`, `R1#wr`, `Building configuration...`, `[OK]`, `R1#`. The R2 configuration commands include: `R2>en`, `R2#conf t`, `R2(config)#interface giga0/2.10`, `R2(config-subif)#standby 10 ip 192.168.10.62`, `R2(config-subif)#standby 10 priority 100`, `R2(config-subif)#`, `R2(config-subif)#interface giga0/2.20`, `R2(config-subif)#standby 20 ip 192.168.20.62`, `R2(config-subif)#standby 20 priority 100`, `R2(config-subif)#interface giga0/2.30`, `R2(config-subif)#standby 30 ip 192.168.30.62`, `R2(config-subif)#standby 30 priority 100`, `R2(config-subif)#`, `R2#`, `%SYS-5-CONFIG_I: Configured from console by console`, `R2#wr`, `Building configuration...`, `[OK]`, `R2#`. Both windows have 'Copy' and 'Paste' buttons at the bottom right.

R1 (Active): Configured HSRP with priority 110 and preempt, making it the active gateway for VLANs 10, 20, and 30.

R2 (Standby): HSRP configuration with priority 100, ensuring redundancy in the event of R1 failure.

3) Dynamic Routing and Redistribution

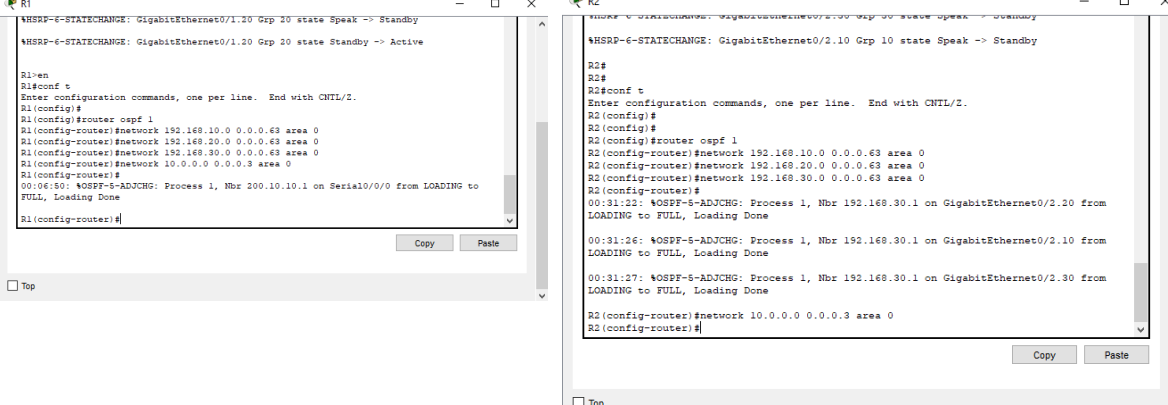
R-DIST:



```
R-DIST
R-DIST>en
R-DIST#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-DIST(config)#
R-DIST(config)#router ospf 1
R-DIST(config-router)#redistribute eigrp 100 subnets
R-DIST(config-router)#redistribute rip subnets
R-DIST(config-router)#network 10.0.0.0 0.0.0.3 area 0
R-DIST(config-router)#network 10.0.0.4 0.0.0.3 area 0
R-DIST(config-router)#network 10.0.10.0 0.0.0.3 area 0
R-DIST(config-router)#exit
R-DIST(config)#router eigrp 100
R-DIST(config-router)#redistribute ospf 1 metric 10000 100 255 1 1500
R-DIST(config-router)#redistribute rip metric 10000 100 255 1 1500
R-DIST(config-router)#network 10.0.20.0 0.0.0.3
R-DIST(config-router)#exit
R-DIST(config)#router rip
R-DIST(config-router)#version 2
R-DIST(config-router)#redistribute ospf 1 metric 5
R-DIST(config-router)#redistribute eigrp 100 metric 5
R-DIST(config-router)#
R-DIST(config-router)#redistribute eigrp 100 metric 5
R-DIST(config-router)#network 10.0.30.0
R-DIST(config-router)#no auto-summary
R-DIST(config-router)#end
R-DIST#
*SYS-5-CONFIG_I: Configured from console by console
R-DIST#
R-DIST#
Building configuration...
[OK]
R-DIST#
```

R-DIST: Redistribution between OSPF (Branch A), EIGRP (Branch B) and RIP (Branch C), ensuring full connectivity between all networks.

R1:R2:

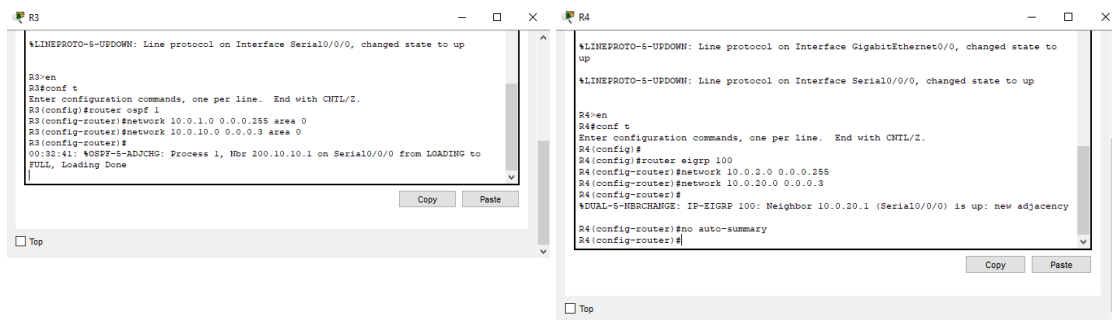


```
R1
%HSRP-6-STATECHANGE: GigabitEthernet0/1.20 Grp 20 state Speak -> Standby
%HSRP-6-STATECHANGE: GigabitEthernet0/1.20 Grp 20 state Standby -> Active
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#router ospf 1
R1(config-router)#network 192.168.10.0 0.0.0.63 area 0
R1(config-router)#network 192.168.20.0 0.0.0.63 area 0
R1(config-router)#network 192.168.30.0 0.0.0.63 area 0
R1(config-router)#network 10.0.0.0 0.0.0.3 area 0
R1(config-router)#
00:06:50: %OSPF-5-ADJCHG: Process 1, Nbr 200.10.10.1 on Serial0/0/0 from LOADING to FULL, Loading Done
R1(config-router)#

R2
%HSRP-6-STATECHANGE: GigabitEthernet0/2.10 Grp 10 state Speak -> Standby
R2#
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#router ospf 1
R2(config-router)#network 192.168.10.0 0.0.0.63 area 0
R2(config-router)#network 192.168.20.0 0.0.0.63 area 0
R2(config-router)#network 192.168.30.0 0.0.0.63 area 0
R2(config-router)#
00:31:22: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on GigabitEthernet0/2.20 from LOADING to FULL, Loading Done
00:31:26: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on GigabitEthernet0/2.10 from LOADING to FULL, Loading Done
00:31:27: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on GigabitEthernet0/2.30 from LOADING to FULL, Loading Done
R2(config-router)#network 10.0.0.0 0.0.0.3 area 0
R2(config-router)#
```

R1 and R2: Enabling OSPF to advertise the Data Center VLANs.

R3:R4:

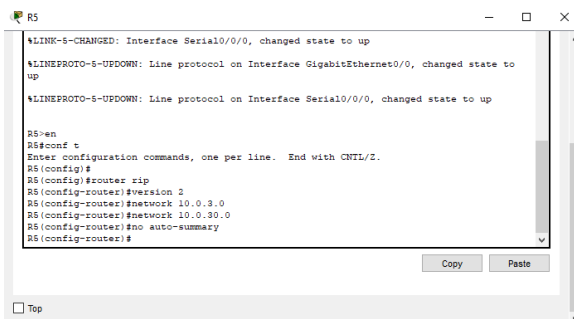


The image shows two terminal windows, R3 and R4, displaying network configuration commands and status messages. Window R3 shows OSPF configuration on Serial10/0/0. Window R4 shows EIGRP configuration on GigabitEthernet0/0 and Serial10/0/0, followed by a status message indicating a new adjacency.

```
R3:
$LINK-5-UPDOWN: Line protocol on Interface Serial10/0/0, changed state to up
R3#
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 10.0.1.0 0.0.0.255 area 0
R3(config-router)#network 10.0.10.0 0.0.0.3 area 0
R3(config-router)#
00:32:41: %OSPF-5-ADJCHG: Process 1, Nbr 200.10.10.1 on Serial10/0/0 from LOADING to FULL, Loading Done

R4:
$LINK-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
$LINK-5-UPDOWN: Line protocol on Interface Serial10/0/0, changed state to up
R4#
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#
R4(config)#router eigrp 100
R4(config-router)#network 10.0.2.0 0.0.0.255
R4(config-router)#network 10.0.20.0 0.0.0.3
R4(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.20.1 (Serial10/0/0) is up: new adjacency
R4(config-router)#no auto-summary
R4(config-router)#
```

R5:



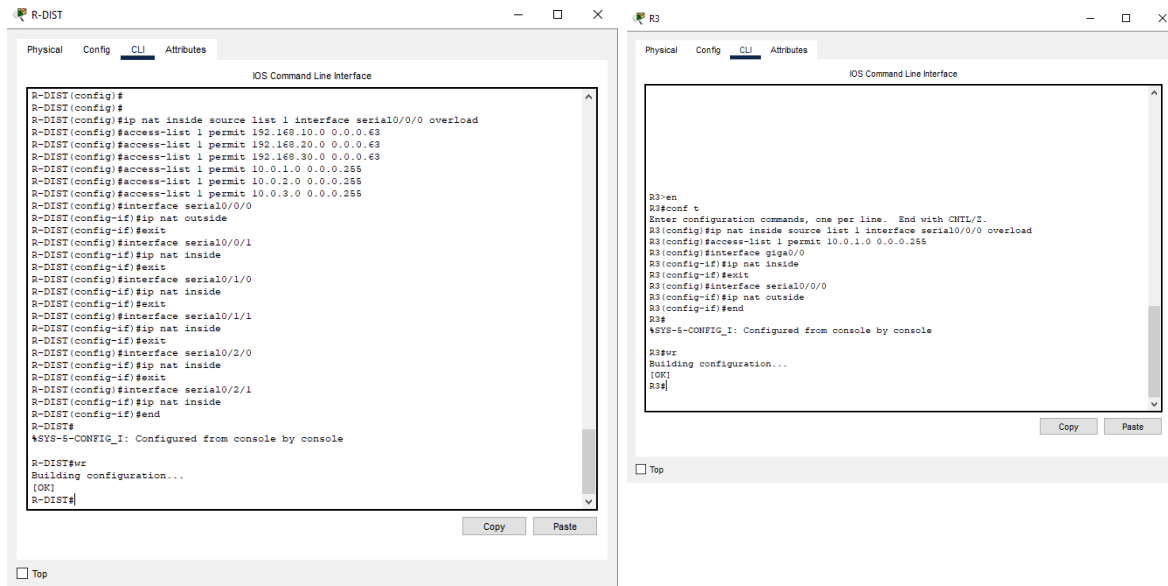
The image shows a terminal window for R5 displaying configuration commands for RIP and OSPF on GigabitEthernet0/0 and Serial10/0/0.

```
R5:
$LINK-5-CHANGED: Interface Serial10/0/0, changed state to up
$LINK-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
$LINK-5-UPDOWN: Line protocol on Interface Serial10/0/0, changed state to up
R5#
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#network 10.0.3.0
R5(config-router)#network 10.0.30.0
R5(config-router)#no auto-summary
R5(config-router)#
```

R3, R4, R5: Activation of OSPF, EIGRP and RIP protocols in their respective branches.

4) NAT (PAT) on all routers

R-DIST: R3:



The image shows two side-by-side screenshots of Cisco IOS Command Line Interface (CLI) windows. The left window is titled 'R-DIST' and the right window is titled 'R3'. Both windows show the configuration of NAT (PAT) on various interfaces. The R-DIST configuration includes setting up an overload on serial0/0/0 and configuring access-lists for internal networks. The R3 configuration shows setting up NAT on gigabitEthernet0/0 and serial0/0/0 interfaces.

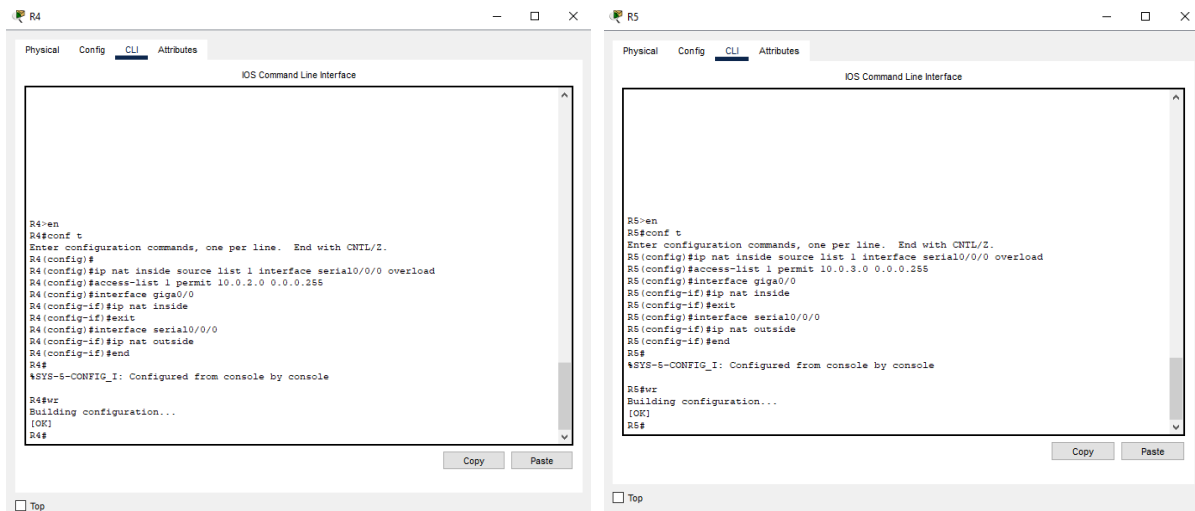
```
R-DIST(config)#
R-DIST(config)#
R-DIST(config)#ip nat inside source list 1 interface serial0/0/0 overload
R-DIST(config)#access-list 1 permit 192.168.10.0 0.0.0.63
R-DIST(config)#access-list 1 permit 192.168.20.0 0.0.0.63
R-DIST(config)#access-list 1 permit 192.168.30.0 0.0.0.63
R-DIST(config)#access-list 1 permit 10.0.1.0 0.0.0.255
R-DIST(config)#access-list 1 permit 10.0.2.0 0.0.0.255
R-DIST(config)#access-list 1 permit 10.0.3.0 0.0.0.255
R-DIST(config)#interface serial0/0/0
R-DIST(config-if)#ip nat outside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/0/1
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/1/0
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/1/1
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/2/0
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/2/1
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#end
R-DIST#
%SYS-5-CONFIG_I: Configured from console by console

R-DIST#vr
Building configuration...
[OK]
R-DIST#
```

```
R3#
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip nat inside source list 1 interface serial0/0/0 overload
R3(config)#access-list 1 permit 10.0.1.0 0.0.0.255
R3(config)#interface giga0/0
R3(config-if)#ip nat inside
R3(config-if)#exit
R3(config)#interface serial0/0/0
R3(config-if)#ip nat outside
R3(config-if)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#vr
Building configuration...
[OK]
R3#
```

R4:R5:



The image shows two side-by-side screenshots of Cisco IOS Command Line Interface (CLI) windows. The left window is titled 'R4' and the right window is titled 'R5'. Both windows show the configuration of NAT (PAT) on various interfaces. The R4 configuration includes setting up an overload on serial0/0/0 and configuring access-lists for internal networks. The R5 configuration shows setting up NAT on gigabitEthernet0/0 and serial0/0/0 interfaces.

```
R4#
R4#en
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#
R4(config)#ip nat inside source list 1 interface serial0/0/0 overload
R4(config)#access-list 1 permit 10.0.2.0 0.0.0.255
R4(config)#interface giga0/0
R4(config-if)#ip nat inside
R4(config-if)#exit
R4(config)#interface serial0/0/0
R4(config-if)#ip nat outside
R4(config-if)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console

R4#vr
Building configuration...
[OK]
R4#
```

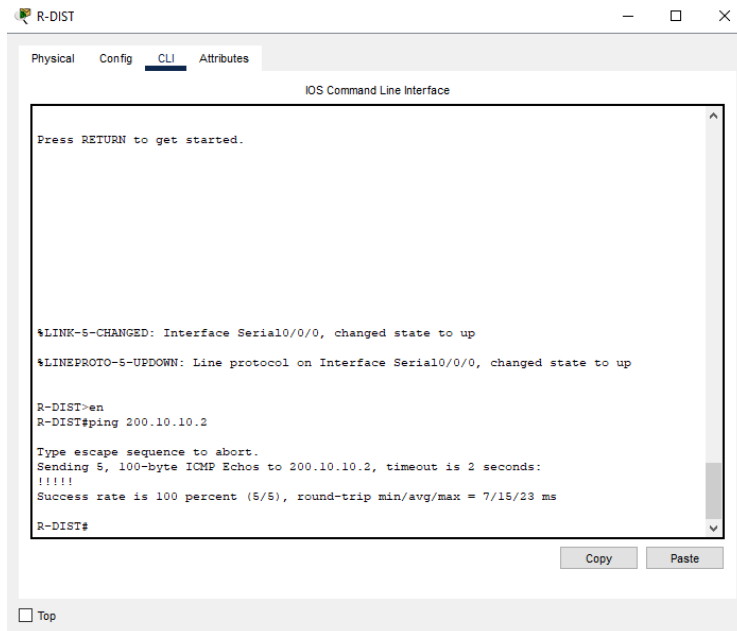
```
R5#
R5#en
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#ip nat inside source list 1 interface serial0/0/0 overload
R5(config)#access-list 1 permit 10.0.3.0 0.0.0.255
R5(config)#interface giga0/0
R5(config-if)#ip nat inside
R5(config-if)#exit
R5(config)#interface serial0/0/0
R5(config-if)#ip nat outside
R5(config-if)#end
R5#
%SYS-5-CONFIG_I: Configured from console by console

R5#vr
Building configuration...
[OK]
R5#
```

R-DIST: NAT overload configuration with ACL 1 defining the authorized internal networks.

R3, R4, R5: Configuration of local NAT to allow Internet access from branch networks.

I now test a ping from the R-DIST router to the ISP's IP address (200.10.10.2), getting a successful response.



```
R-DIST
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R-DIST>en
R-DIST#ping 200.10.10.2

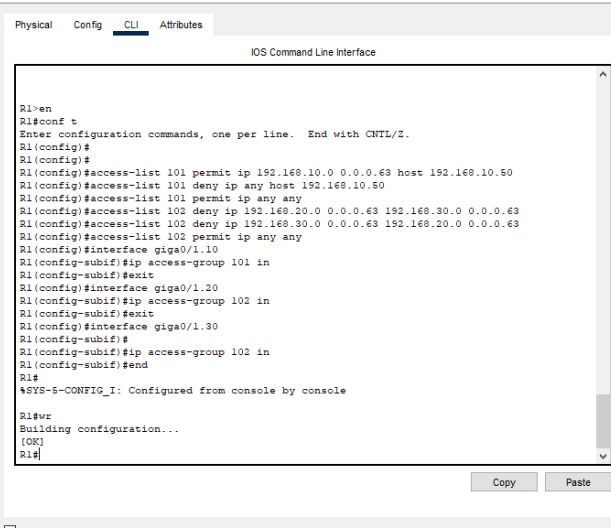
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 7/15/23 ms

R-DIST#
```

The PCs' communication will be tested later, after the DHCP service is configured.

5) ACLs – Access Control Lists

A1:

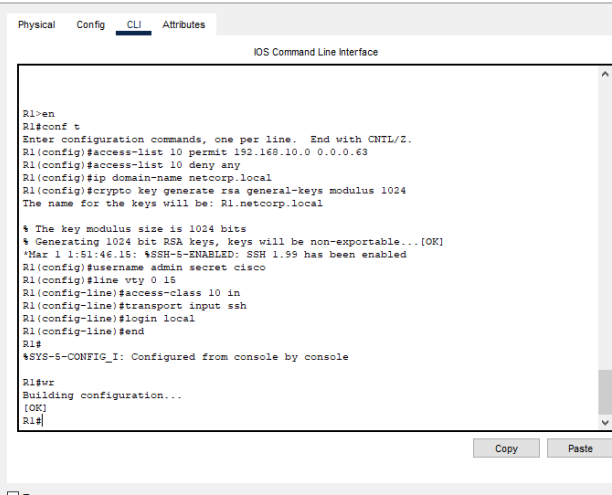


```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#access-list 101 permit ip 192.168.10.0 0.0.0.63 host 192.168.10.50
R1(config)#access-list 101 deny ip any host 192.168.10.50
R1(config)#access-list 102 deny ip 192.168.20.0 0.0.0.63 192.168.30.0 0.0.0.63
R1(config)#access-list 102 deny ip 192.168.30.0 0.0.0.63 192.168.20.0 0.0.0.63
R1(config)#access-list 102 permit ip any any
R1(config)#interface giga0/1.10
R1(config-subif)#ip access-group 101 in
R1(config-subif)#exit
R1(config)#interface giga0/1.20
R1(config-subif)#ip access-group 102 in
R1(config-subif)#exit
R1(config)#interface giga0/1.30
R1(config-subif)#ip access-group 102 in
R1(config-subif)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#wr
Building configuration...
[OK]
R1#
```

A1: Enforcing ACLs to restrict access to SRV1 (VLAN 10 only) and block communication between VLAN 20 and VLAN 30.

All devices:



```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.63
R1(config)#access-list 10 deny any
R1(config)#ip domain-name netcorp.local
R1(config)#crypto key generate rsa general-keys modulus 1024
The name for the keys will be: R1.netcorp.local

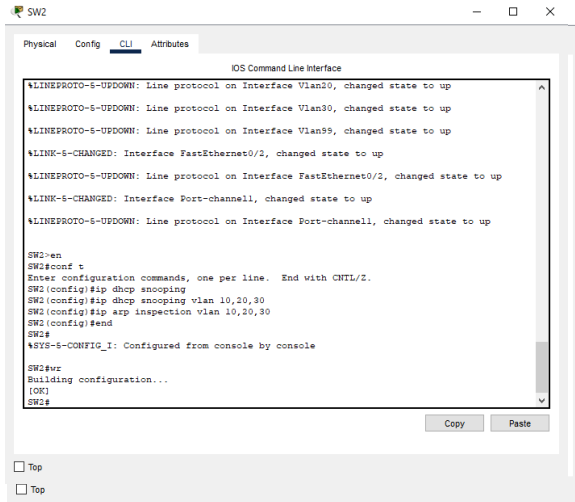
% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:51:46.16: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#username admin secret cisco
R1(config)#line vty 0 15
R1(config-line)#access-class 10 in
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#wr
Building configuration...
[OK]
R1#
```

All devices: Creation of ACL 10 to restrict SSH to VLAN 10 and enable SSH with Telnet disablement.

6) Layer 2 Security

SW1:SW2:



```
SW2
Physical Config CLI Attributes
IOS Command Line Interface

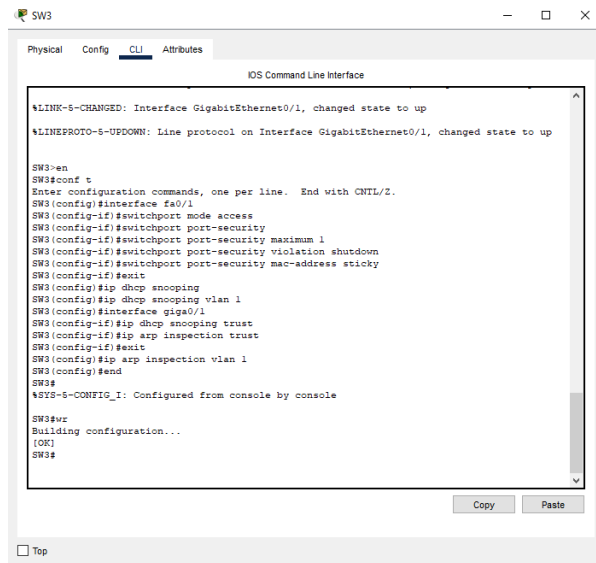
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

SW2>en
SW2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#ip dhcp snooping
SW2(config)#ip dhcp snooping vlan 10,20,30
SW2(config)#ip arp inspection vlan 10,20,30
SW2(config)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console

SW2#wr
Building configuration...
[OK]
SW2#
```

SW1 and SW2: Activation of DHCP Snooping and DAI, with definition of trusted ports (R1, R2, SRV1).

SW3, SW4 and SW5:



```
SW3
Physical Config CLI Attributes
IOS Command Line Interface

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW3>en
SW3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW3(config)#interface fa0/1
SW3(config-if)#switchport mode access
SW3(config-if)#switchport port-security
SW3(config-if)#switchport port-security maximum 1
SW3(config-if)#switchport port-security violation shutdown
SW3(config-if)#switchport port-security mac-address sticky
SW3(config-if)#exit
SW3(config)#ip dhcp snooping
SW3(config)#ip dhcp snooping vlan 1
SW3(config)#interface giga0/1
SW3(config-if)#ip dhcp snooping trust
SW3(config-if)#ip arp inspection trust
SW3(config-if)#exit
SW3(config)#ip arp inspection vlan 1
SW3(config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

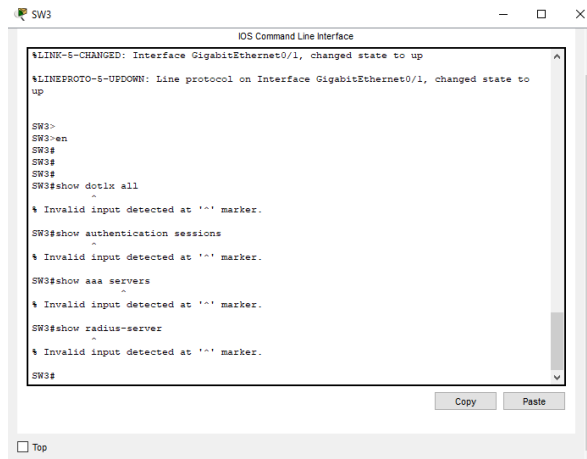
SW3#wr
Building configuration...
[OK]
SW3#
```

Same configuration for all 3, changing only the interfaces used if they are different.

SW3, SW4, SW5: Configuration of Port-Security on PC ports and activation of DHCP Snooping and DAI on branch office VLANs.

7) AAA, IEEE 802.1X and SSH authentication

During the implementation of Exercise 7, several attempts were made to configure 802.1X with RADIUS authentication, following the procedures used in the classes, however, even after configuring everything, the fundamental verification commands were not available:



```
SW3
IOS Command Line Interface
ALINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
SW3>
SW3>en
SW3#
SW3#
SW3#show dot1x all
^
* Invalid input detected at '^' marker.
SW3#show authentication sessions
^
* Invalid input detected at '^' marker.
SW3#show aaa servers
^
* Invalid input detected at '^' marker.
SW3#show radius-server
^
* Invalid input detected at '^' marker.
SW3#
```

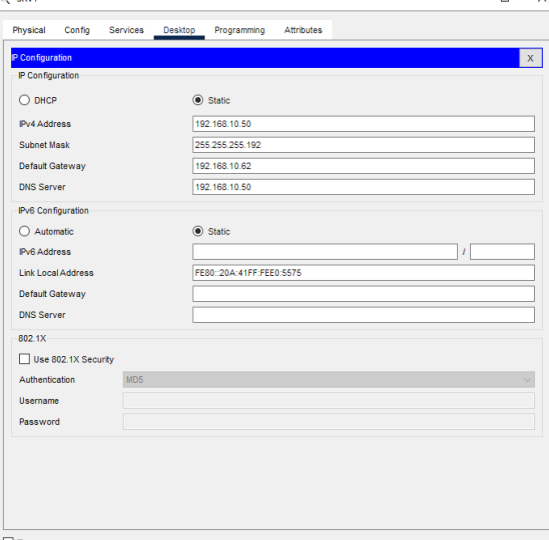
The port remains blocked and DHCP fails, even with the correct credentials on the PC.

It was not possible to determine with certainty if this failure results from a limitation of the simulator (version 8.2.2.0400) or some error in the configuration on my part, although apparently correct, it is not accepted by Packet Tracer. What is certain is that in this release, both 802.1X and RADIUS do not behave as expected and it was not possible to make them functional.

Because of this I decided to do this exercise without using 802.1X or RADIUS, to fulfill the objective of the statement a functional alternative solution was implemented, consisting of:

- Port-Security with MAC sticky for door access control.
- Local authentication with user1/cinel and admin/cisco users.
- Active SSH on all equipment with ACL restricted to VLAN 10.
- DHCP relay (ip helper-address) for dynamic assignment of IPs.

SRV1:



SRV1

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

Pv4 Address: 192.168.10.50

Subnet Mask: 255.255.255.192

Default Gateway: 192.168.10.62

DNS Server: 192.168.10.50

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::20A:41FF:FE00:5575

Default Gateway:

DNS Server:

802.1X

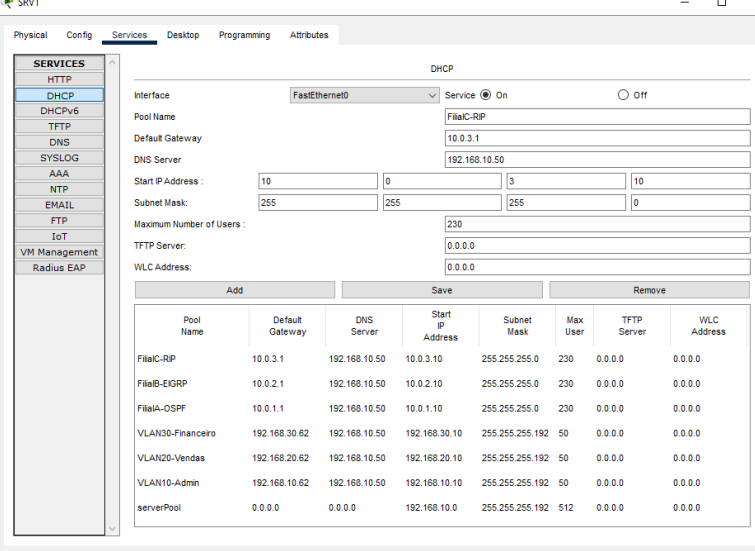
☐ Use 802.1X Security

Authentication: MDS

Username:

Password:

☐ Top



SRV1

Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: FiliaC-RP

Default Gateway: 10.0.3.1

DNS Server: 192.168.10.50

Start IP Address: 10 0 3 10

Subnet Mask: 255 255 255 0

Maximum Number of Users: 230

TFTP Server: 0.0.0.0

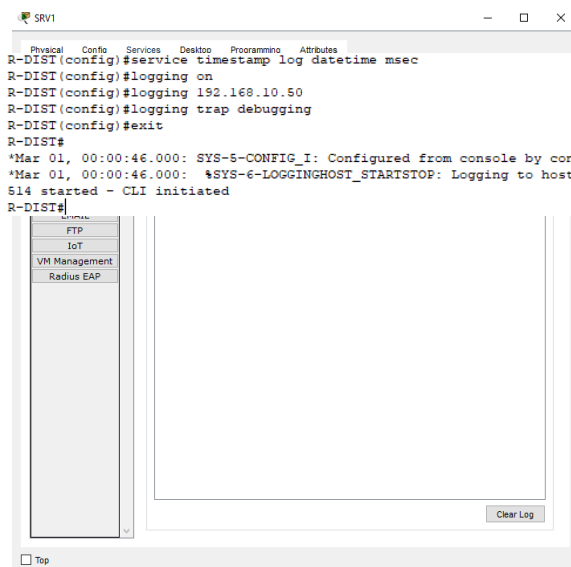
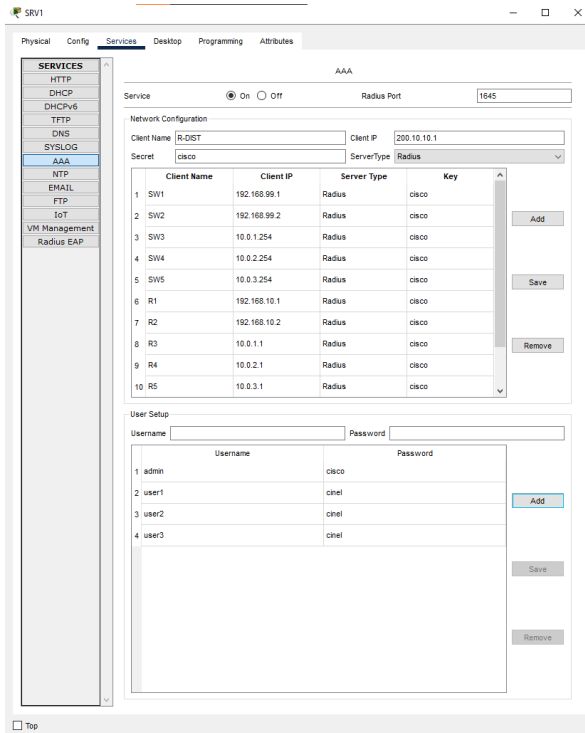
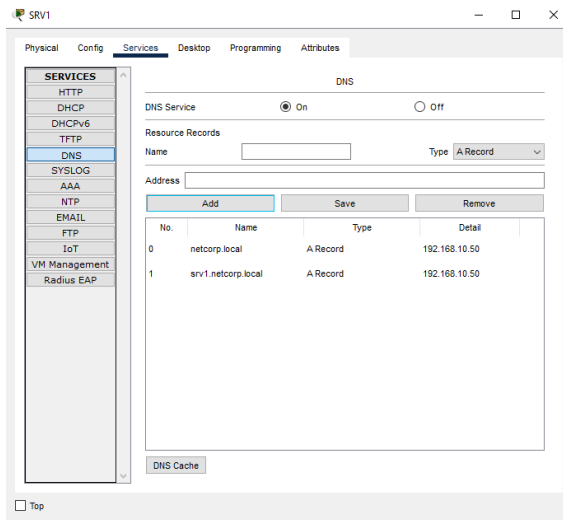
WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
FiliaC-RP	10.0.3.1	192.168.10.50	10.0.3.10	255.255.255.0	230	0.0.0.0	0.0.0.0
FiliaB-EXRP	10.0.2.1	192.168.10.50	10.0.2.10	255.255.255.0	230	0.0.0.0	0.0.0.0
FiliaA-OSPF	10.0.1.1	192.168.10.50	10.0.1.10	255.255.255.0	230	0.0.0.0	0.0.0.0
VLAN30-Financeiro	192.168.30.62	192.168.10.50	192.168.30.10	255.255.255.192	50	0.0.0.0	0.0.0.0
VLAN20-Vendas	192.168.20.62	192.168.10.50	192.168.20.10	255.255.255.192	50	0.0.0.0	0.0.0.0
VLAN10-Admin	192.168.10.62	192.168.10.50	192.168.10.10	255.255.255.192	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.10.0	255.255.255.192	512	0.0.0.0	0.0.0.0

☐ Top

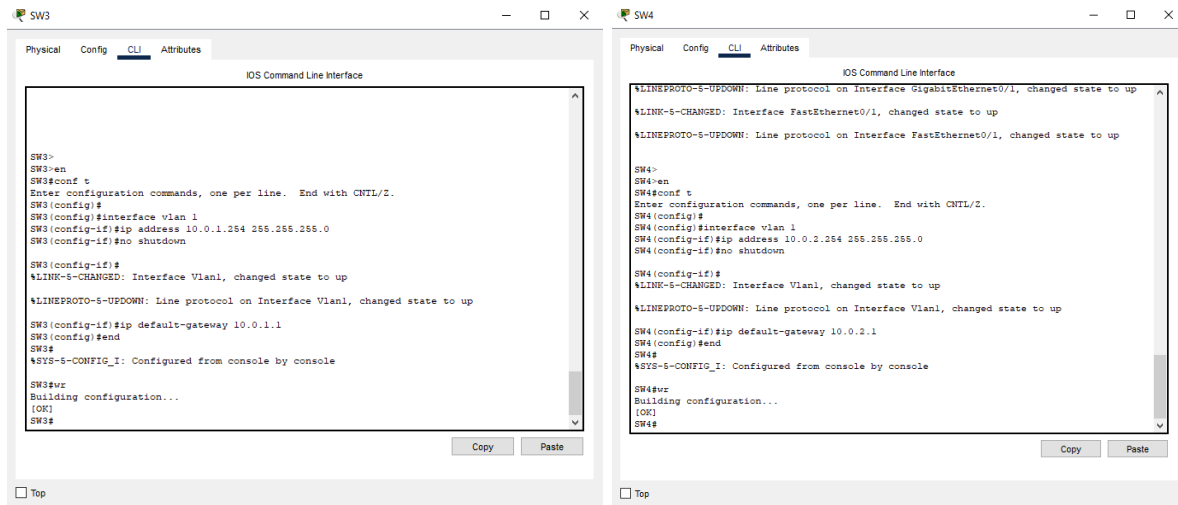
Despite being ON I ended up not using AAA:



R-DIST:

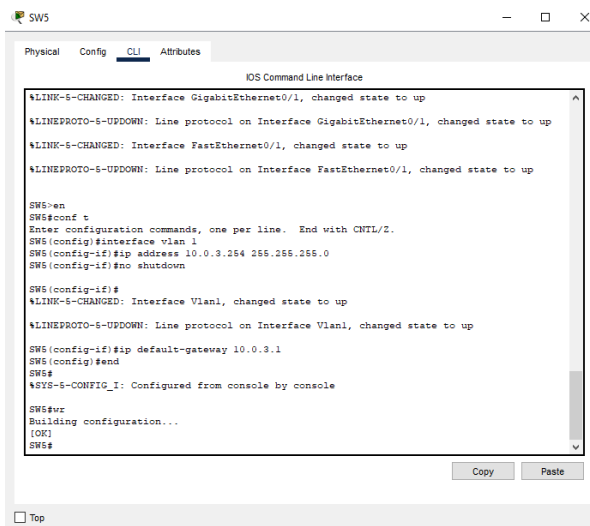
SRV1: Configuration of the server with fixed IP and activation of DHCP and DNS services for all networks.

SW3:SW4:



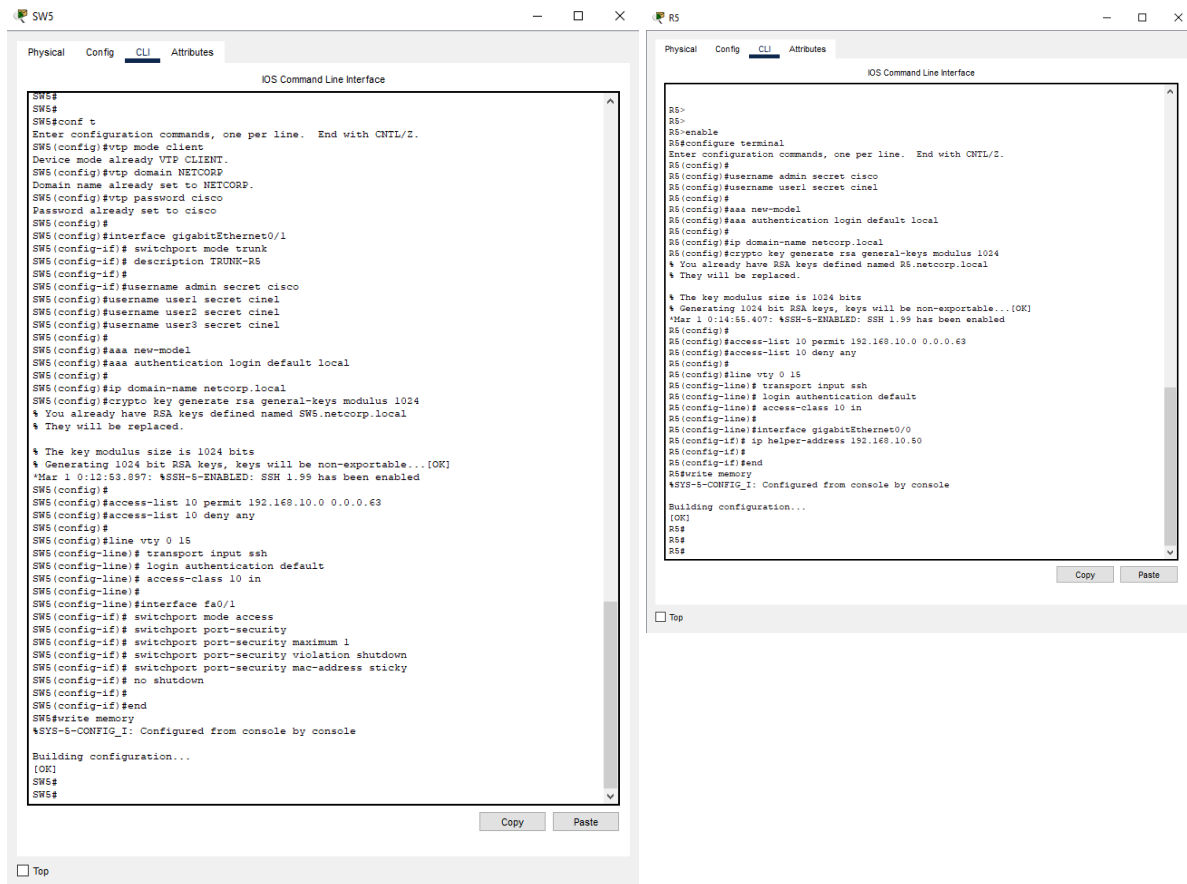
The image shows two side-by-side screenshots of network switch configuration windows. The left window is for SW3 and the right window is for SW4. Both windows show the 'CLI' tab with the 'IOS Command Line Interface' text. The configuration commands for SW3 are: SW3> SW3#en SW3#conf t Enter configuration commands, one per line. End with CNTL/Z. SW3(config)# SW3(config)#interface vlan 1 SW3(config-if)#ip address 10.0.1.254 255.255.255.0 SW3(config-if)#no shutdown SW3(config-if)# %LINK-6-CHANGED: Interface Vlan1, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up SW3(config-if)#ip default-gateway 10.0.1.1 SW3(config)#end SW3# %SYS-5-CONFIG_I: Configured from console by console SW3# SW3# Building configuration... (OK) SW3# The configuration commands for SW4 are: SW4> SW4#en SW4#conf t Enter configuration commands, one per line. End with CNTL/Z. SW4(config)# SW4(config)#interface vlan 1 SW4(config-if)#ip address 10.0.2.254 255.255.255.0 SW4(config-if)#no shutdown SW4(config-if)# %LINK-6-CHANGED: Interface Vlan1, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up SW4(config-if)#ip default-gateway 10.0.2.1 SW4(config)#end SW4# %SYS-5-CONFIG_I: Configured from console by console SW4# SW4# Building configuration... (OK) SW4#

SW5:



The image shows a screenshot of a network switch configuration window for SW5. The window shows the 'CLI' tab with the 'IOS Command Line Interface' text. The configuration commands for SW5 are: SW5> SW5#en SW5#conf t Enter configuration commands, one per line. End with CNTL/Z. SW5(config)# SW5(config)#interface vlan 1 SW5(config-if)#ip address 10.0.3.254 255.255.255.0 SW5(config-if)#no shutdown SW5(config-if)# %LINK-6-CHANGED: Interface Vlan1, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up SW5(config-if)#ip default-gateway 10.0.3.1 SW5(config)#end SW5# %SYS-5-CONFIG_I: Configured from console by console SW5# SW5# Building configuration... (OK) SW5#

SW3, SW4, SW5: R3, R4, R5:



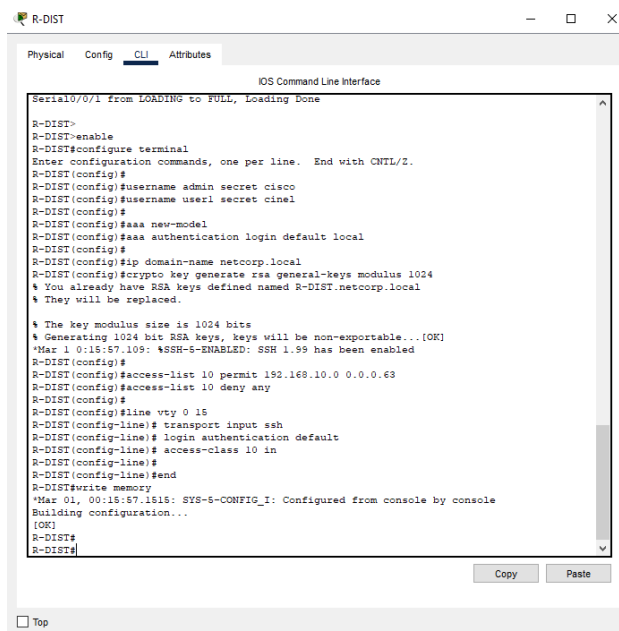
```
SW5#
SW5#
SW5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW5(config)#vtp mode client
Device mode already VTP CLIENT.
SW5(config)#vtp domain NETCORP
Domain name already set to NETCORP.
SW5(config)#vtp password cisco
Password already set to cisco
SW5(config)#
SW5(config)#interface gigabitEthernet0/1
SW5(config-if)# switchport mode trunk
SW5(config-if)# description TRUNK-R5
SW5(config-if)#
SW5(config-if)#username admin secret cisco
SW5(config-if)#username user1 secret cinel
SW5(config-if)#username user2 secret cinel
SW5(config-if)#username user3 secret cinel
SW5(config-if)#
SW5(config)#aaa new-model
SW5(config)#aaa authentication login default local
SW5(config)#
SW5(config)#ip domain-name netcorp.local
SW5(config)#crypto key generate rsa general-keys modulus 1024
% You already have RSA keys defined named SW5.netcorp.local
% They will be replaced.

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 0:12:53.897: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW5(config)#
SW5(config)#access-list 10 permit 192.168.10.0 0.0.0.63
SW5(config)#access-list 10 deny any
SW5(config)#
SW5(config)#line vty 0 15
SW5(config-line)# transport input ssh
SW5(config-line)# login authentication default
SW5(config-line)# access-class 10 in
SW5(config-line)#
SW5(config-line)#interface fa0/1
SW5(config-if)# switchport mode access
SW5(config-if)# switchport port-security
SW5(config-if)# switchport port-security maximum 1
SW5(config-if)# switchport port-security violation shutdown
SW5(config-if)# switchport port-security mac-address sticky
SW5(config-if)# no shutdown
SW5(config-if)#
SW5(config-if)#end
SW5#write memory
*SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
SW5#
SW5#
```

```
R5>
R5>
R5>enable
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#
R5(config)#username admin secret cisco
R5(config)#username user1 secret cinel
R5(config)#
R5(config)#aaa new-model
R5(config)#aaa authentication login default local
R5(config)#
R5(config)#ip domain-name netcorp.local
R5(config)#crypto key generate rsa general-keys modulus 1024
% You already have RSA keys defined named R5.netcorp.local
% They will be replaced.

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 0:14:55.407: %SSH-5-ENABLED: SSH 1.99 has been enabled
R5(config)#
R5(config)#access-list 10 permit 192.168.10.0 0.0.0.63
R5(config)#access-list 10 deny any
R5(config)#
R5(config)#line vty 0 15
R5(config-line)# transport input ssh
R5(config-line)# login authentication default
R5(config-line)# access-class 10 in
R5(config-line)#
R5(config-line)#interface gigabitEthernet0/0
R5(config-if)# ip helper-address 192.168.10.50
R5(config-if)#
R5(config-if)#end
R5#write memory
*SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
R5#
R5#
R5#
```

R-DIST, R1, R2:

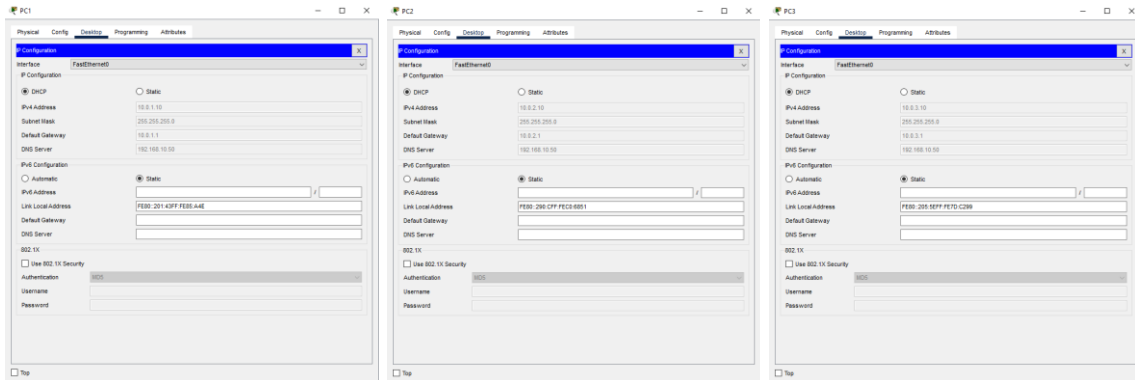


```
R-DIST>
R-DIST>
R-DIST>enable
R-DIST#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R-DIST(config)#
R-DIST(config)#username admin secret cisco
R-DIST(config)#username user1 secret cinel
R-DIST(config)#
R-DIST(config)#aaa new-model
R-DIST(config)#aaa authentication login default local
R-DIST(config)#
R-DIST(config)#ip domain-name netcorp.local
R-DIST(config)#crypto key generate rsa general-keys modulus 1024
% You already have RSA keys defined named R-DIST.netcorp.local
% They will be replaced.

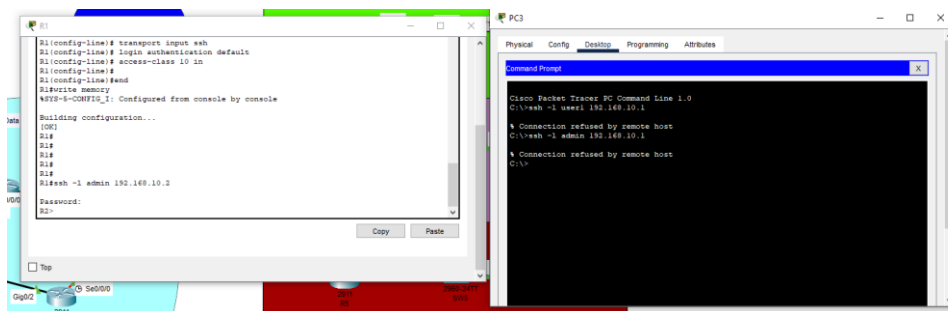
% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 0:15:57.109: %SSH-5-ENABLED: SSH 1.99 has been enabled
R-DIST(config)#
R-DIST(config)#access-list 10 permit 192.168.10.0 0.0.0.63
R-DIST(config)#access-list 10 deny any
R-DIST(config)#
R-DIST(config)#line vty 0 15
R-DIST(config-line)# transport input ssh
R-DIST(config-line)# login authentication default
R-DIST(config-line)# access-class 10 in
R-DIST(config-line)#
R-DIST(config-line)#end
R-DIST#write memory
*Mar 01, 00:15:57.1616: SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
R-DIST#
R-DIST#
```

Switches (SW3, SW4, SW5): Creation of local users, activation of AAA with local authentication, configuration of SSH with ACL and implementation of Port-Security.

Routers: Configuration of local authentication and SSH on all routers, with ip helper-address on branch routers for DHCP relay.

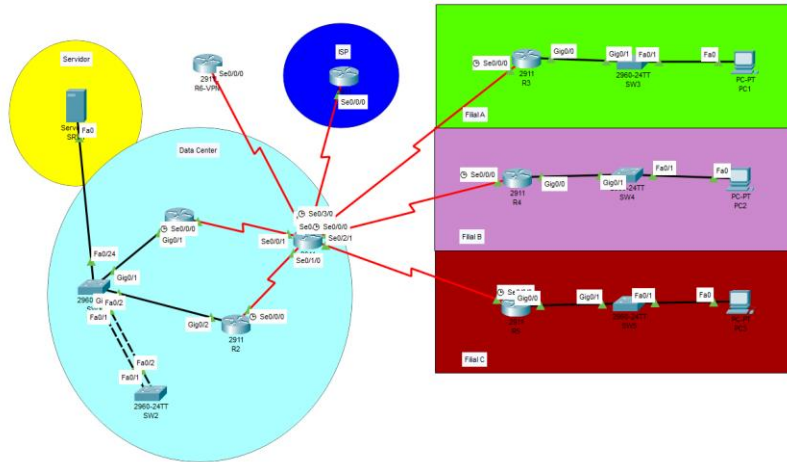


Only devices on vlan 10 can ssh, acl blocks the rest.



8) VPN – Virtual Private Network

For this exercise I'm going to add a router with a different network and connect it to R-DIST:
R6-VPN (se0/0/0) → (se0/3/0) R-DIST



Before setting up R-DIST I first add securityk9:

R-DIST# license boot module c2900 technology-package securityk9
R-DIST# reload

R-DIST: R6-VPN:

```

R-DIST#
R-DIST#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-DIST(config)#interface Serial0/3/0
R-DIST(config-if)#description LINK-R6
R-DIST(config-if)#ip address 200.10.20.1 255.255.255.252
R-DIST(config-if)#clock rate 64000
This command applies only to DCE interfaces
R-DIST(config-if)#no shutdown
R-DIST(config-if)#exit
R-DIST(config)#interface Tunnel0
R-DIST(config-if)#
Mar 01, 00:03:07.033: %LINK-5-CHANGED: Interface Tunnel0, changed state to up
R-DIST(config-if)#description VPN-TUNNEL-GRE-PADA-R6
% Invalid input detected at ... marker.
R-DIST(config-if)#ip address 10.0.99.1 255.255.255.252
R-DIST(config-if)#tunnel source Serial0/3/0
R-DIST(config-if)#tunnel destination 200.10.20.2
R-DIST(config-if)#tunnel mode gre ip
R-DIST(config-if)#exit
R-DIST(config)#ip route 172.16.1.0 255.255.255.0 10.0.99.3
R-DIST(config)#end
R-DIST#
Mar 01, 00:03:40.033: SYS-5-CONFIG_1: Configured from console by console
R-DIST#
Building configuration...
[OK]
R-DIST#

```

```

R6-VPN#
Enter configuration commands, one per line. End with CNTL/Z.
R6-VPN(config)#hostname R6-VPN
R6-VPN(config)#interface Serial0/0/0
R6-VPN(config-if)#description LINK-R-DIST
R6-VPN(config-if)#ip address 200.10.20.2 255.255.255.252
R6-VPN(config-if)#no shutdown
R6-VPN(config-if)#exit
R6-VPN(config)#interface Loopback0
R6-VPN(config-if)#
Mar 01, 00:03:07.033: %LINK-5-CHANGED: Interface Loopback0, changed state to up
R6-VPN(config-if)#
R6-VPN(config)#interface Tunnel0
R6-VPN(config-if)#
Mar 01, 00:03:07.033: %LINK-5-CHANGED: Interface Tunnel0, changed state to up
R6-VPN(config-if)#description VPN-TUNNEL-GRE-PADA-R-DIST
% Invalid input detected at ... marker.
R6-VPN(config-if)#ip address 10.0.99.2 255.255.255.252
R6-VPN(config-if)#tunnel source Serial0/0/0
R6-VPN(config-if)#tunnel destination 200.10.20.1
R6-VPN(config-if)#exit
R6-VPN(config)#interface Tunnel0
R6-VPN(config-if)#
Mar 01, 00:03:07.033: %LINK-5-CHANGED: Interface Tunnel0, changed state to up
R6-VPN(config-if)#tunnel mode gre ip
R6-VPN(config-if)#exit
R6-VPN(config)#ip route 192.168.10.0 255.255.255.0 10.0.99.1
R6-VPN(config)#ip route 192.168.20.0 255.255.255.0 10.0.99.1
R6-VPN(config)#ip route 192.168.30.0 255.255.255.0 10.0.99.1
R6-VPN(config)#ip route 0.0.0.0 0.0.0.0 200.10.20.1
R6-VPN(config)#end
R6-VPN#
SYS-5-CONFIG_1: Configured from console by console
R6-VPN#
Building configuration...
[OK]
R6-VPN#

```

R-DIST and R6: Creation of a GRE tunnel between R-DIST and the remote R6 router, with static routes to allow communication between the internal networks and the VPN client.

9) Testing and Validation

Performing ping tests between branches, checking routing tables, analyzing NAT, querying ACLs and Port-Security, and validating SSH with created users:

Test ping between branches:

```
R3#ping 10.0.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 25/33/39 ms

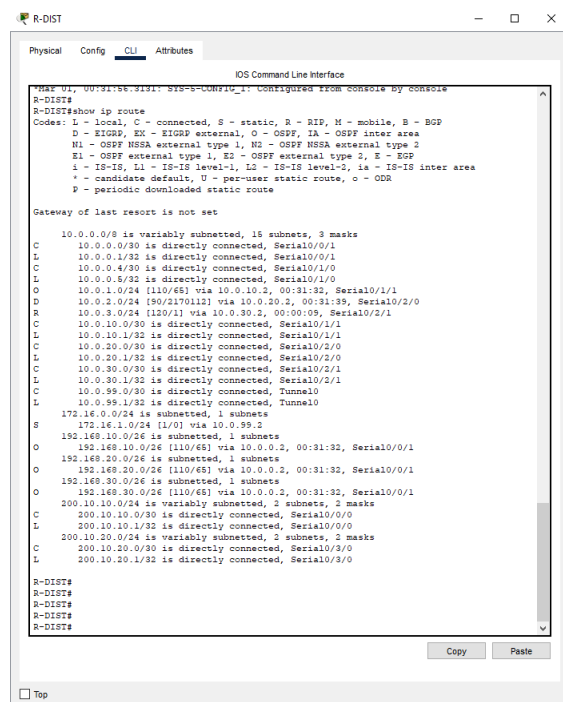
R4#ping 10.0.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 21/33/39 ms

R5#ping 10.0.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 21/34/40 ms
```

Check route redistribution:



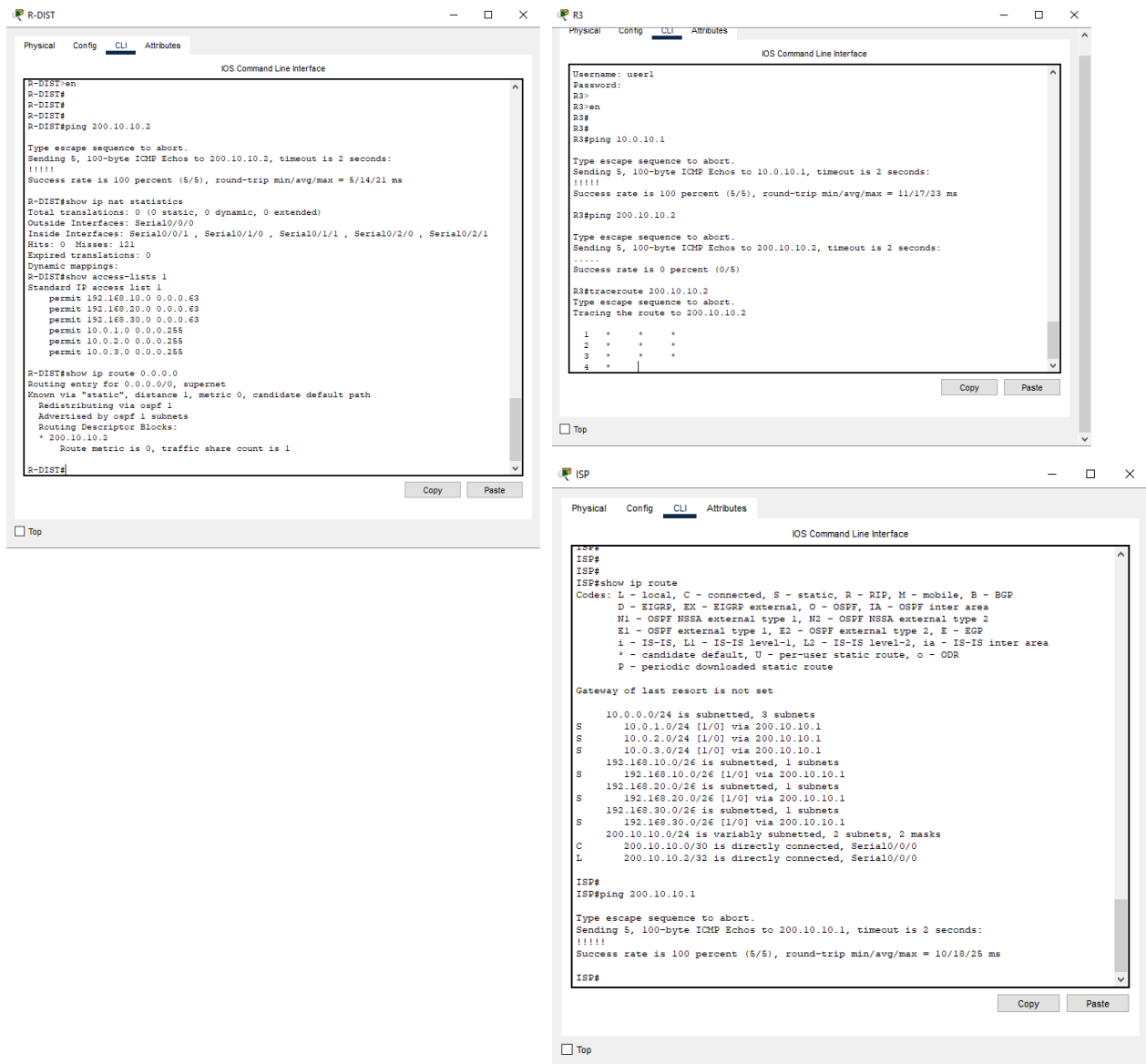
```
R-DIST
Physical Config CLI Attributes
IOS Command Line Interface
R-DIST#
R-DIST#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        O - OSPF, EX - OSPF external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, Ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 15 subnets, 3 masks
C    10.0.0.0/30 is directly connected, Serial0/0/1
L    10.0.0.1/32 is directly connected, Serial0/0/1
C    10.0.0.4/30 is directly connected, Serial0/1/0
L    10.0.0.5/32 is directly connected, Serial0/1/0
O    10.0.1.0/24 [110/65] via 10.0.10.2, 00:31:32, Serial0/1/1
D    10.0.2.0/24 [90/2170112] via 10.0.20.2, 00:31:39, Serial0/2/0
R    10.0.3.0/24 [130/1] via 10.0.30.2, 00:00:09, Serial0/2/1
C    10.0.10.0/30 is directly connected, Serial0/1/1
L    10.0.10.1/32 is directly connected, Serial0/1/1
C    10.0.20.0/30 is directly connected, Serial0/2/0
L    10.0.20.1/32 is directly connected, Serial0/2/0
C    10.0.30.0/30 is directly connected, Serial0/2/1
L    10.0.30.1/32 is directly connected, Serial0/2/1
C    10.0.99.0/30 is directly connected, Tunnel0
L    10.0.99.1/32 is directly connected, Tunnel0
S    172.16.0.0/24 is subnetted, 1 subnets
    S    172.16.1.0/24 [1/0] via 10.0.99.2
O    192.168.10.0/26 is subnetted, 1 subnets
    O    192.168.10.0/26 [110/65] via 10.0.0.2, 00:31:32, Serial0/0/1
O    192.168.20.0/26 is subnetted, 1 subnets
    O    192.168.20.0/26 [110/65] via 10.0.0.2, 00:31:32, Serial0/0/1
O    192.168.30.0/26 is subnetted, 1 subnets
    O    192.168.30.0/26 [110/65] via 10.0.0.2, 00:31:32, Serial0/0/1
C    200.10.10.0/30 is directly connected, Serial0/0/0
L    200.10.10.1/32 is directly connected, Serial0/0/0
C    200.10.20.0/24 is variably subnetted, 2 subnets, 2 masks
    C    200.10.20.0/30 is directly connected, Serial0/3/0
    L    200.10.20.1/32 is directly connected, Serial0/3/0

R-DIST#
R-DIST#
R-DIST#
R-DIST#
R-DIST#
```

Validate NAT:



```
R-DIST>en
R-DIST#
R-DIST#
R-DIST#
R-DIST#ping 200.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.10.10.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 5/14/21 ms

R-DIST#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: Serial0/0/0
Inside Interfaces: Serial0/0/1, Serial0/1/0, Serial0/1/1, Serial0/2/0, Serial0/2/1
Hits: 0 Misses: 121
Expired translations: 0
Dynamic mappings:
R-DIST#show access-lists 1
Standard IP access list 1
 permit 192.168.10.0 0.0.0.63
 permit 192.168.20.0 0.0.0.63
 permit 192.168.30.0 0.0.0.63
 permit 10.0.1.0 0.0.0.255
 permit 10.0.2.0 0.0.0.255
 permit 10.0.3.0 0.0.0.255

R-DIST#show ip route 0.0.0.0
Routing entry for 0.0.0.0/0, supernet
Known via "static", distance 1, metric 0, candidate default path
Redistributing via ospf 1
Advertised by ospf 1 subnets
Routing Descriptor Blocks:
* 200.10.10.2
  Route metric is 0, traffic share count is 1

R-DIST#

R3>
R3>
R3>en
R3#
R3#ping 10.0.10.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.10.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 11/17/23 ms

R3#ping 200.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.10.10.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R3#traceroute 200.10.10.2
Type escape sequence to abort.
Tracing the route to 200.10.10.2
 1 * * *
 2 * * *
 3 * * *
 4 * * *

ISP>
ISP#
ISP#
ISP#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/24 is subnetted, 3 subnets
   S   10.0.1.0/24 [1/0] via 200.10.10.1
   S   10.0.2.0/24 [1/0] via 200.10.10.1
   S   10.0.3.0/24 [1/0] via 200.10.10.1
 192.168.10.0/26 is subnetted, 1 subnets
   S   192.168.10.0/26 [1/0] via 200.10.10.1
 192.168.20.0/26 is subnetted, 1 subnets
   S   192.168.20.0/26 [1/0] via 200.10.10.1
 192.168.30.0/26 is subnetted, 1 subnets
   S   192.168.30.0/26 [1/0] via 200.10.10.1
 200.10.10.0/24 is variably subnetted, 2 subnets, 2 masks
   C   200.10.10.0/30 is directly connected, Serial0/0/0
   L   200.10.10.2/32 is directly connected, Serial0/0/0

ISP#
ISP#ping 200.10.10.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.10.10.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/18/25 ms

ISP#
```

The NAT configuration has been done: R-DIST has default route, ACL 1 with all internal networks, and the traffic arrives at NAT (Misses: 101). R3 reaches R-DIST, and the ISP has return routes and responds to pings from R-DIST.

However, the ping from R3 to the ISP fails and the traceroute shows that the packet does not advance. The show ip nat translations command remains empty, indicating that the translation does not occur.

I believe that in a real network, with physical equipment, the configuration would behave differently.

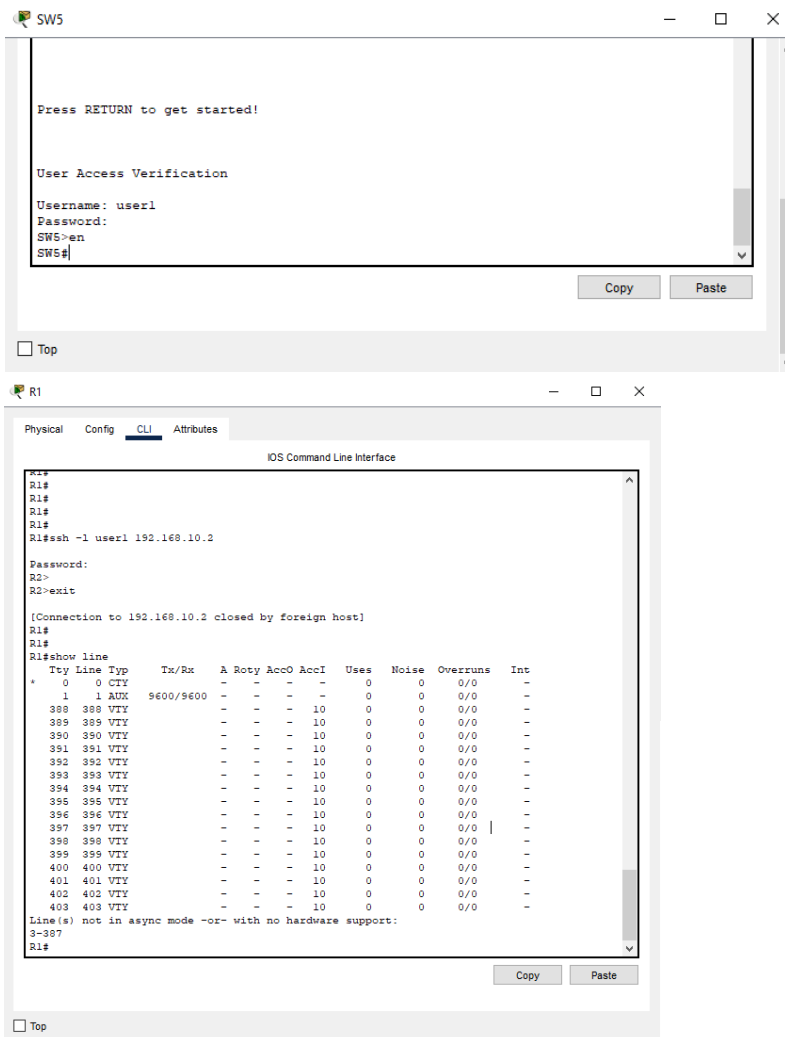
Confirm ACLs and port security:

```
R1
User Access Verification
Username:
Username: admin
Password:
R1>
R1>en
R1#
R1#
R1#
R1#show access-lists
Extended IP access list 101
 10 permit ip 192.168.10.0 0.0.0.63 host 192.168.10.50
 20 deny ip any host 192.168.10.50
 30 permit ip any any (4830 match(es))
Extended IP access list 102
 10 deny ip 192.168.20.0 0.0.0.63 192.168.30.0 0.0.0.63
 20 deny ip 192.168.30.0 0.0.0.63 192.168.20.0 0.0.0.63
 30 permit ip any any (7135 match(es))
Standard IP access list 10
 10 permit 192.168.10.0 0.0.0.63
 20 deny any (48 match(es))
R1#
```

```
SW5
Physical Config CLI Attributes
IOS Command Line Interface

User Access Verification
Username: user1
Password:
SW5>
SW5>en
SW5#
SW5#show port-security interface fa0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 0005.5E7D.C299:1
Security Violation Count : 0
SW5#
```

Test AAA and SSH:



The screenshot displays two windows from a network simulation. The top window, titled 'SW5', shows a user access verification process where 'user1' is successfully authenticated. The bottom window, titled 'R1', shows the configuration of an AAA server and the execution of an SSH command. The command 'R1#ssh -l user1 192.168.10.2' is entered, and the output shows a successful connection to the host at 192.168.10.2. Below the command output, the 'show line' command is executed, displaying a table of line configurations.

```
Press RETURN to get started!

User Access Verification

Username: user1
Password:
SW5>en
SW5#
```

Top

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1#
R1#
R1#
R1#
R1#ssh -l user1 192.168.10.2
Password:
R2>
R2>exit
[Connection to 192.168.10.2 closed by foreign host]
R1#
R1#show line
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
Tty Line Typ Tx/Rx A Roty AccO AccI Uses Noise Overruns Int
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
* 0 0 CTY - - - - 0 0 0/0 -
1 1 AUX 9600/9600 - - - - 0 0 0/0 -
388 388 VTY - - - - 10 0 0 0/0 -
389 389 VTY - - - - 10 0 0 0/0 -
390 390 VTY - - - - 10 0 0 0/0 -
391 391 VTY - - - - 10 0 0 0/0 -
392 392 VTY - - - - 10 0 0 0/0 -
393 393 VTY - - - - 10 0 0 0/0 -
394 394 VTY - - - - 10 0 0 0/0 -
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396 396 VTY - - - - 10 0 0 0/0 -
397 397 VTY - - - - 10 0 0 0/0 -
398 398 VTY - - - - 10 0 0 0/0 -
399 399 VTY - - - - 10 0 0 0/0 -
400 400 VTY - - - - 10 0 0 0/0 -
401 401 VTY - - - - 10 0 0 0/0 -
402 402 VTY - - - - 10 0 0 0/0 -
403 403 VTY - - - - 10 0 0 0/0 -
Line(s) not in async mode -or- with no hardware support:
3-397
R1#
```

Top

Bibliography

Sources of information used:

- Recordings of the classes,
- Work previously done,
- Notes.

Conclusion

The NETCORP network, when implemented, fulfilled most of the proposed requirements. The infrastructure is functional, secure and redundant, with full communication between branches and the Data Center.

During the project, some technical limitations emerged, namely 802.1X and NAT, which did not work as expected.

I couldn't resolve such errors because I didn't realize if they were errors in my configuration or bugs in the Packet Tracer version (Version 8.2.2.0400).

Whenever possible, alternatives were implemented to ensure compliance with the objectives of the statement.

This work allowed to consolidate the knowledge acquired throughout the UFCD, constituting an enriching learning experience.